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*** START OF THE PROJECT GUTENBERG EBOOK MED SERVICE

MED SERVICE

BY MURRAY LEINSTER

Illustrated by van Dongen

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I

"The probability of unfavorable consequences cannot be zero in any action of common life, but the probability increases by a very high power as a series of actions is lengthened. The effect of moral considerations, in conduct, may be stated to be a mathematically verifiable reduction in the number of unfavorable possible chance happenings. Of course, whether this process is termed the intelligent use of probability, or ethics, or piety, makes no difference in the fact. It is the method by which unfavorable chance happenings are made least probable. Arbitrary actions such as we call criminal cannot ever be justified by mathematics. For example ..."

Probability and Human Conduct
Fitzgerald

Calhoun lay in his bunk and read Fitzgerald on "Probability and Human Conduct" as the little Med Ship floated in overdrive. In overdrive travel there is nothing to do but pass the time away. Murgatroyd, the *tormal*, slept curled up in a ball in one corner of the small ship's cabin. His tail was meticulously curled about his nose. The ship's lights burned steadily. There were those small random noises which have to be provided to keep a man sane in the dead stillness of a ship traveling at thirty times the speed of light. Calhoun turned a page and yawned.

Something stirred somewhere. There was a click, and a taped voice said:

"When the tone sounds, break-out will be five seconds off."

A metronomic ticking, grave and deliberate, resounded in the stillness. Calhoun heaved himself up from the bunk and marked his place in the book. He moved to and seated himself in the control chair and fastened the safety belt. He said:

"Murgatroyd! Hark, hark the lark in Heaven's something-or-other doth sing. Wake up and comb your whiskers. We're getting there."

Murgatroyd opened one eye and saw Calhoun in the pilot's chair. He uncurled himself and padded to a place where there was something to grab hold of. He regarded Calhoun with bright eyes.

"*BONG!*" said the tape. It counted down. "Five—four—three—two—one—"

It stopped. The ship popped out of overdrive. The sensation was unmistakable. Calhoun's stomach seemed to turn over twice, and he had a sickish feeling of spiraling dizzily in what was somehow a cone. He swallowed. Murgatroyd made gulping noises. Outside, everything changed.

The sun Maris blazed silently in emptiness off to port. The Cetis star-cluster was astern, and the light by which it could be seen had traveled for many years to reach here, though Calhoun had left Med Headquarters only three weeks before. The third planet of Maris swung splendidly in its orbit. Calhoun checked, and nodded in satisfaction. He spoke over his shoulder to Murgatroyd.

"We're here, all right."

"*Chee!*" shrilled Murgatroyd.

He uncoiled his tail from about a cabinet-handle and hopped up to look at the vision-screen. What he saw, of course, meant nothing to him. But all *tormals* imitate the actions of human beings, as parrots imitate their speech. He blinked wisely at the screen and turned his eyes to Calhoun.

"It's Maris III," Calhoun told him, "and pretty close. It's a colony of Dettra Two. One city was reported started two Earth-years ago. It should just about be colonized now."

"*Chee-chee!*" shrilled Murgatroyd.

"So get out of the way," commanded Calhoun. "We'll make our approach and I'll tell 'em we're here."

He made a standard approach on interplanetary drive. Naturally, it was a long process. But after some hours he flipped over the call-switch and made the usual identification and landing request.

"Med Ship *Aesclipus Twenty* to ground," he said into the transmitter. "Requesting co-ordinates for landing. Our mass is fifty tons. Repeat, five-oh tons. Purpose of landing, planetary health inspection."

He relaxed. This job ought to be purest routine. There was a landing grid in the spaceport city on Maris III. From its control room instructions should be sent, indicating a position some five planetary diameters or farther out from the surface of that world. Calhoun's little ship should repair to that spot. The giant landing grid should then reach out its specialized force-field and lock onto the ship, and then bring it gently but irresistibly down to ground. Then Calhoun, representing Med Service, should confer gravely with planetary authorities about public health conditions on Maris III.

It was not to be expected that anything important would turn up. Calhoun would deliver full details of recent advances in the progress of medicine. These might already have reached Maris III in the ordinary course of commerce, but he would make sure. He might—but it was unlikely—learn of some novelty worked out here. In any case, within three days he should return to the small Med Ship, the landing grid should heave it firmly heavenward to not less than five planetary diameters distance, and there release it. And Calhoun and Murgatroyd and the Med Ship should flick into overdrive and speed back toward Headquarters, from whence they had come.

Right now, Calhoun waited for an answer to his landing call. But he regarded the vast disk of the nearby planet.

"By the map," he observed to Murgatroyd, "the city ought to be on the shore of that bay somewhere near the terminus. Close to the sunset line."

His call was answered. A voice said incredulously on the space-phone speaker:

"*What? What's that? What's that you say?*"

"Med Ship *Aesclipus Twenty*," Calhoun repeated patiently. "Requesting coordinates for landing. Our mass is fifty tons. Repeat, five-oh tons. Purpose of landing, planetary health inspection."

The voice said more incredulously still:

"*A Med Ship? Holy—*" By the change of sound, the man down on the planet had turned away from the microphone. "*Hey! Listen to this—*"

There was abrupt silence. Calhoun raised his eyebrows. He drummed on the control desk before him.

There was a long pause. A very long pause. Then a new voice came on the space phone, up from the ground:

"You up aloft there! Identify yourself!"

Calhoun said very politely:

"This is Med Ship *Aesclipus Twenty*. I would like to come to ground. Purpose of landing, health inspection."

"Wait," said the voice from the planet. It sounded strained.

A murmuring sounded, transmitted from fifty thousand miles away. Then there was a click. The transmitter down below had cut off. Calhoun raised his eyebrows again. This was not according to routine. Not at all! The Med Service was badly overworked and understaffed. The resources of interplanetary services were always apt to be stretched to their utmost, because there could be no galactic government as such. Some thousands of occupied planets, the closest of them light-years apart—or weeks of traveling—couldn't hold elections or have political parties for the simple reason that travel even in overdrive was too slow. They could only have service organizations whose authority depended on the consent of the people served, and whose support had to be gathered when and as it was possible.

But the Med Service was admittedly important. The local Sector Headquarters was in the Cetus cluster. It was a sort of interstellar clinic, with additions. It gathered and disseminated the results of experience in health and medicine among some thousands of colony-worlds, and from time to time it made contact with other Headquarters carrying on the same work elsewhere. It admittedly took fifty years for a new technique in gene-selection to cross the so-far-occupied part of the galaxy, but it was a three-year voyage in overdrive to cover the same distance direct. And the Med Service was worth while. There was no problem of human ecological adjustment it had so far been unable to solve, and there were some dozens of planets whose human colonies owed their existence to it. There was nowhere—nowhere at all—that a Med Ship was not welcomed on its errand from Headquarters.

"Aground there!" said Calhoun sharply. "What's the matter? Are you landing me or not?"

There was no answer. Then, suddenly, every sound-producing device in the ship abruptly emitted a hoarse and monstrous noise. The lights flashed up and circuit-breakers cut them off. The nearby-object horn squawked. The hull-temperature warning squealed. The ship's internal gravity-field tugged horribly for an instant and went off. Every device within the ship designed to notify emergency clanged or shrieked or roared or screamed. There was a momentary bedlam.

It lasted for part of a second only. Then everything stopped. There was no weight within the ship, and there were no lights, and there was dead silence, and Murgatroyd made whimpering sounds in the darkness.

Calhoun thought absurdly to himself, *"According to the book, this is an unfavorable chance consequence of something or other."* But it was more than an unfavorable chance occurrence. It was an intentional and drastic and possibly a deadly one.

"Somebody's acting up," said Calhoun measuredly, in the blackness. "What's the matter with them?"

He flipped the screen switch to bring back vision of what was outside. The vision screens of a ship are very carefully fused against over-load burnouts, because there is nothing in all the cosmos quite as helpless and foredoomed as a ship which is blind in the emptiness of space. But the screens did not light again. They couldn't. The cutouts hadn't worked in time.

Calhoun's scalp crawled. But as his eyes adjusted, he saw the palely fluorescent handles of switches and doors. They hadn't been made fluorescent in expectation of an emergency like this, of course, but they would help a great deal. He knew what had happened. It couldn't be but one thing—a landing-grid field clamped on the fifty-ton Med Ship with the power needed to grasp and land a twenty-thousand-ton liner. At that strength it would paralyze every instrument and blow every cut-off. It could not be accident. The reception of the news of his identity, the repeated request that he identify himself, and then the demand that he wait—This murderous performance was deliberate.

"Maybe," said Calhoun in the inky-black cabin, "as a Med Ship our arrival is an unfavorable chance consequence of something—or the unfavorableness is—and somebody means to keep us from happening. It looks like it."

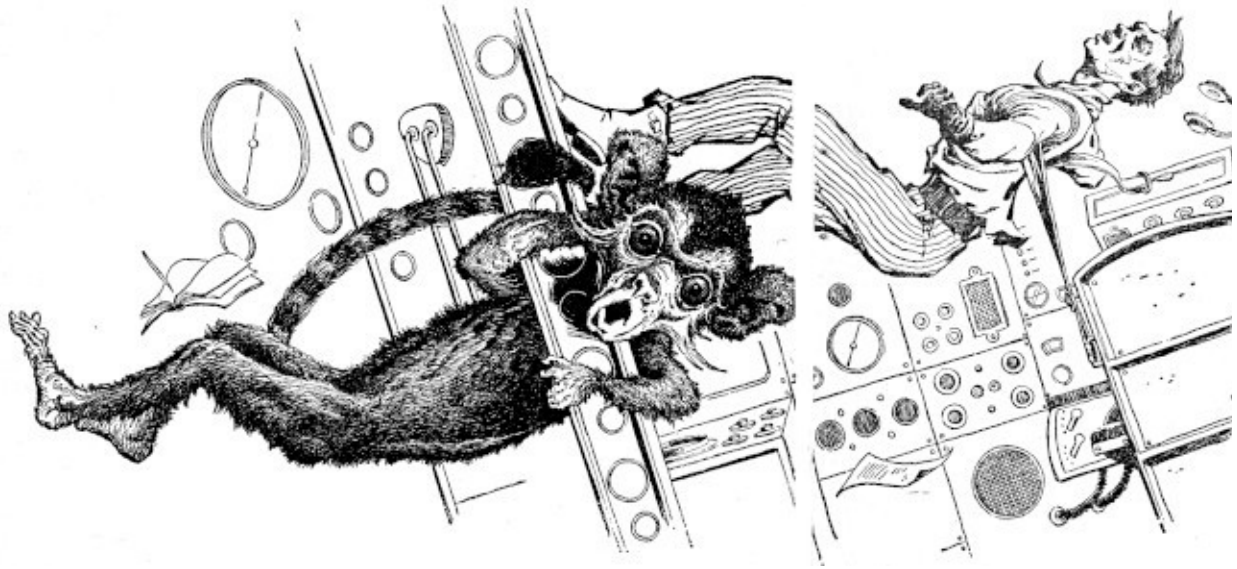
Murgatroyd whimpered.

"And I think," added Calhoun coldly, "that somebody may need a swift kick in the negative feedback!"

He released himself from the safety belt and dived across the cabin in which there was now no weight at all. In the blackness he opened a cabinet door. What he did inside was customarily done by a man wearing thick insulating gloves, in the landing grid back at Headquarters. He threw certain switches which would allow the discharge of the power-storage cells which worked the Med Ship's overdrive. Monstrous quantities of energy were required to put even a fifty-ton ship into overdrive, and monstrous amounts were returned when it came out. The power amounted to ounces of pure, raw energy, and as a safety-precaution such amounts were normally put into the Duhanne cells only just before a Med Ship's launching, and drained out again on its return. But now, Calhoun threw switches which made a rather incredible amount of power available for dumping into the landing-grid field about him—if necessary.

He floated back to the control chair.

The ship lurched—violently. It was being moved by the grid field without any gentleness at all. Calhoun's hands barely grasped the back of his pilot's chair before the jerk came, and it almost tore them free. He just missed being flung against the back wall of the cabin by the applied acceleration. But he was a long way out from the planet. He was at the end of a lever fifty thousand miles long. For that lever to be used to shake him too brutally would require special adjustments. But somebody was making them. The jerk reversed directions. He was flung savagely against the chair to which he'd been clinging. He struggled. Another yank, in another direction. Another one still. It flung him violently into the chair.



Behind him, Murgatroyd squealed angrily as he went hurtling across the cabin. He grabbed for holding-places with all four paws and his tail.

Another shake. Calhoun had barely clipped the safety belt fast before a furious jolt nearly flung him out of it again to crash against the cabin ceiling. Yet another vicious surge of acceleration. He scrabbled for the controls. The yanks and plungings of the ship increased intolerably. He was nauseated. Once he was thrust so furiously into the control chair that he was on the verge of blacking out, and then the direction of thrust was changed to the exact opposite so that the blood rushing to his head seemed about to explode it. His arms flailed out of control. He became dazed. But when his hands were flung against the control board, despite their bruising he tried to cling to the control knobs, and each time he threw them over. Practically all his circuits were blown, but there was one—

His numbing fingers threw it. There was a roar, so fierce that it seemed an explosion. He'd reached the switch which made effective the discharge-circuit of his Duhanne cells. He'd thrown it. It was designed to let the little ship's overdrive power-reserve flow into storage at Headquarters on return from duty. Now, though, it poured into the landing-grid field outside. It amounted to hundreds of millions of kilowatt hours, delivered in the fraction of a second. There was the smell of ozone. The sound was like a thunder clap.

But abruptly there was a strange and incredible peace. The lights came on waveringly as his shaking fingers restored the circuit-breakers. Murgatroyd

shrilled indignantly, clinging desperately to an instrument rack. But the vision screens did not light again. Calhoun swore. Swiftly, he threw more circuit-restorers. The nearest-object indicator told of the presence of Maris III at forty-odd thousand miles. The hull-temperature indicator was up some fifty-six degrees. The internal-gravity field came on, faintly, and then built up to normal. But the screens would not light. They were permanently dead. Calhoun raged for seconds. Then he got hold of himself.

"*Chee-chee-chee!*" chattered Murgatroyd desperately. "*Chee-chee!*"

"Shut up!" growled Calhoun. "Some bright lad aground thought up a new way to commit murder. Damned near got away with it, too! He figured he'd shake us to death like a dog does a rat, only he was using a landing-grid field to do it with! Right now, I hope I fried him!"

But it was not likely. Such quantities of power as are used to handle twenty-thousand-ton space liners are not controlled direct, but by relays. The power Calhoun had flung into the grid field should have blown out the grid's transformers with a spectacular display of fireworks, but it was hardly probable it had gotten back to the individual at the controls.

"But I suspect," observed Calhoun vengefully, "that he'll consider this business an unfavorable occurrence! Somebody'll twist his tail, too, either for trying what he did or for not getting away with it! Only, as a matter of pure precaution—"

His expression changed suddenly. He'd been trying not to think of the consequences of having no sight of the cosmos outside the ship. Now he remembered the electron telescope. It had not been in circuit, so it could not have been burned out like his vision screens. He switched it on. A star field appeared over his head.

"*Chee-chee!*" cried Murgatroyd hysterically.

Calhoun glanced at him. The jerking of the ship had shifted the instruments in the rack to which Murgatroyd clung. Clipped into place though they were, they'd caught Murgatroyd's tail and pinched it tightly.

"You'll have to wait," snapped Calhoun. "Right now I've got to make us look like a successful accident. Otherwise whoever tried to spread us all over the cabin walls will try something else!"

The Med Ship flung through space in whatever direction and at whatever velocity it had possessed when the grid field blew. Calhoun shifted the electron-telescope field and simultaneously threw on the emergency-rocket controls. There was a growling of the pencil-thin, high-velocity blasts. There was a surging of the ship.

"No straight-line stuff," Calhoun reminded himself.

He swung the ship into a dizzy spiral, as if innumerable things had been torn or battered loose in the ship and its rockets had come on of themselves. Painstakingly, he jettisoned in one explosive burst all the stored waste of his journey which could not be disposed of while in overdrive. To any space-scanning instrument on the ground, it would look like something detonating violently inside the ship.

"Now—"

The planet Maris III swung across the electron telescope's field. It looked hideously near—but that was the telescope's magnification. Yet Calhoun sweated. He looked at the nearest-object dial for reassurance. The planet was nearer by a thousand miles.

"Hah!" said Calhoun.

He changed the ship's spiral course. He changed it again. He abruptly reversed the direction of its turn. Adequate training in space-combat might have helped plot an evasion-course, but it might have been recognizable. Nobody could anticipate his maneuvers now, though. He adjusted the telescope next time the planet swept across its field, and flipped on the photorecorder. Then he pulled out of the spiral, whirled the ship until the city was covered by the telescope, and ran the recorder as long as he dared keep a straight course. Then he swooped toward the planet in a crazy, twisting fall with erratic intermissions, and made a final lunatic dash almost parallel to the planet's surface.

At five hundred miles, he unshielded the ports which of necessity had to be kept covered in clear space. There was a sky which was vividly bright with stars. There was a vast blackness off to starboard which was the night side of the planet.

He went down. At four hundred miles the outside pressure indicator wavered away from its pin. He used it like a pitot-tube recording, doing sums in his

head to figure the static pressure that should exist at this height, to compare with the dynamic pressure produced by his velocity through the near hard vacuum. The pressure should have been substantially zero. He swung the ship end-for-end and killed velocity to bring the pressure-indication down. The ship descended. Two hundred miles. He saw the thin bright line of sunshine at the limb of the planet. Down to one hundred. He cut the rockets and let the ship fall silently, swinging it nose up.

At ten miles he listened for man-made radiation. There was nothing in the electromagnetic spectrum but the crackling of static in an electric storm which might be a thousand miles away. At five miles height the nearest-object indicator, near the bottom of its scale, wavered in a fashion to prove that he was still moving laterally across mountainous country. He swung the ship and killed that velocity, too.

At two miles he used the rockets for deceleration. The pencil-thin flame reached down for an incredible distance. By naked-eye observation out a port he tilted the fiercely-roaring, swiftly-falling ship until hillsides and forests underneath him ceased to move. By that time he was very low indeed.



He reached ground on a mountainside which was lighted by the blue-white flame of the rocket-blast. He chose an area in which the tree tops were almost flat, indicating something like a plateau underneath. Murgatroyd was practically frantic by this time because of his capture and the pinching of his tail, but Calhoun could not spare time to release him. He let the ship down gently, gently, trying to descend in an absolutely vertical line.

If he didn't do it perfectly, he came very close. The ship settled into what was practically a burned-away tunnel among monstrous trees. The high-velocity slender flame did not splash when it reached ground. It penetrated. It burned a hole for itself through humus and clay and bedrock. When the ship touched and settled, there was boiling molten stone some sixty feet underground, but there was a small scratching sound as it came to rest. A flame-amputated tree-limb rubbed tentatively against the hull.

Calhoun turned off the rockets. The ship swayed slightly and there were crunching noises. Then it was still on its landing fins.

"Now," said Calhoun, "I can take care of you, Murgatroyd."

He flicked on the switches of the exterior microphones—much more sensitive than human ears. The radiation-detectors were still in action. They reported only the cracklings of the distant storm.

But the microphones brought in the moaning of wind over nearby mountaintops, and the almost deafening susurrus of rustling leaves. Underneath these noises there was a bedlam of other natural sounds. There were chirpings and hootings and squeaks, and the gruntings made by native animal life. These sounds had a singularly peaceful quality. When Calhoun toned them down to be no more than background-noises, they suggested the sort of concert of night-creatures which to men has always seemed an indication of purest tranquility.

Presently Calhoun looked at the pictures the photorecorder had taken while the telescope's field swept over the city. It was the colony-city reported to have been begun two years before, to receive colonists from Dettra Two. It was the city of the landing grid which had tried to destroy the Med Ship as a dog kills a rat—by shaking it to fragments, some fifty thousand miles in space. It was the city which had made Calhoun land with his vision-plates blinded, that had made him pretend his ship was internally a wreck: which had drained his power-reserves of some hundreds of millions of kilowatt hours of energy. It was the city which had made his return to Med Headquarters impossible.

He inspected the telescopic pictures. They were very clear. They showed the city with astonishing detail. There was a lacy pattern of highways, with their medallions of multiple-dwelling units. There were the lavish park areas between the buildings of this planetary capital. There was the landing grid itself—a half-mile-high structure of steel girders, a full mile in diameter.

But there were no vehicles on the highways. There were no specks on the crossing bridges to indicate people on foot. There were no copters on the building roofs, nor were there objects in mid-air to tell of air traffic.

The city was either deserted or it had never been occupied. But it was absolutely intact. The structures were perfect. There was no indication of

past panic or disaster, and even the highways had not been overgrown by vegetation. But it was empty—or else it was dead.

But somebody in it had tried very ferociously and with singular effectiveness to try to destroy the Med Ship.

Because it was a Med Ship.

Calhoun raised his eyebrows and looked at Murgatroyd.

"Why is all this?" he asked. "Have you any ideas?"

"*Chee!*" shrilled Murgatroyd.

II

"The purpose of a contemplated human action is always the attainment of a desired subjective experience. But a subjective experience is desired both in terms of intensity, and in terms of duration. For an individual the temptingness of different degrees of intensity-of-experience is readily computed. However, the temptingness of different durations is equally necessary for the computation of the probability of a given individual performing a given action. This modification of desirability by expectable duration depends on the individual's time-sense: its acuity and its accuracy. Measurements of time-sense—"

Probability and Human Conduct
Fitzgerald

Two days later Calhoun found a cultivated field and a dead man, but before that he found only bewilderment. Before leaving the Med Ship, he very carefully monitored all over again the entire radiation spectrum for man-made signals. There were no communications in the air of Maris III—which on the face of it was proof that the planet was uninhabited. But the ship's external microphones picked up a rocket roar in mid-morning of the day after Calhoun's landing. By the time the sound reached the ground, of course, the rocket itself was far below the horizon; but Calhoun saw the faint

white trail of its passage against the blue of the sky. The fact that he saw it, in daytime, was proof that it was within the atmosphere. Which, in turn, said that the rocket was taking photographs from high altitude for signs of the crater the Med Ship should have made in an uncontrolled landing.

The fact of search proved that the planet was inhabited, and the silence of the radio spectrum said that it wasn't. The absence of traffic in the city said that it was dead or empty, but there were people there because they'd answered Calhoun's hail, and tried to kill him when he identified himself. But nobody would want to destroy a Med Ship except to prevent a health inspection, and nobody would want to prevent an inspection unless there was a situation aground that the Med Service ought to know about. But there should not possibly be such a situation.

There was no logical explanation for such a series of contradictions. Civilized men acted this way or that. There could only be civilized men here. They acted neither this way nor that. Therefore—the confusion began all over again.

Calhoun dictated an account of events to date into the emergency responder in the ship. If a search-call came from space, the responder would broadcast this data and Calhoun's intended action. He carefully shut off all other operating circuits so the ship couldn't be found by their radiation. He equipped himself for travel, and he and Murgatroyd left the ship. Obviously, he headed toward the city where whatever was wrong was centered.

Travel on foot was unaccustomed, but not difficult. The vegetation was semi-familiar. Maris III was an Earth-type planet and circled a Sol-type sun, and given similar conditions of gravity, air, sunlight, and temperature-range, similar organisms should develop. There would be room, for example, for low-growing ground-cover plants and there would also be advantages to height. There would be some equivalent of grasses, and there would be the equivalent of trees, with intermediate forms having in-between habits of growth. Similar reasoning would apply to animal life. There would be parallel ecological niches for animals to fill, and animals would adapt to fill them.

Maris III was not, then, an "unearthly" environment. It was much more like an unfamiliar part of a known planet than a new world altogether. But there were some oddities. An herbivorous creature without legs which squirmed

like a snake. It lived in holes. A pigeon-sized creature whose wings were modified, gossamer-thin scales with iridescent colorings. There were creatures which seemed to live in lunatic association, and Calhoun was irritably curious to know if they were really symbiotes or only unrecognizable forms of the same organism, like the terrestrial male and female firefly-glowworm.

But he was heading for the city. He couldn't spare time to biologize. On his first day's journey he looked for food to save the rations he carried. Murgatroyd was handy, here. The little *tormal* had his place in human society. He was friendly, and he was passionately imitative of human beings, and he had a definite psychology of his own. But he was useful, too. When Calhoun strode through the forests which had such curiously un-leaflike foliage, Murgatroyd strode grandly with him, imitating his walk. From time to time he dropped to all four paws to investigate something. He invariably caught up with Calhoun within seconds.

Once Calhoun saw him interestedly bite a tiny bit out of a most unpromising looking shrub-stalk. He savored its flavor, and then swallowed it. Calhoun took note of the plant and cut off a section. He bound it to the skin of his arm up near the elbow. Hours later there was no allergic reaction, so he tasted it. It was almost familiar. It had the flavor of a bracken-shoot, mingled with a fruity taste. It would be a green bulk-food like spinach or asparagus, filling but without much substance.

Later, Murgatroyd carefully examined a luscious-seeming fruit which grew low enough for him to pluck. He sniffed it closely and drew back. Calhoun noted that plant, too. Murgatroyd's tribe was bred at Headquarters for some highly valuable qualities. One was a very sensitive stomach—but it was only one. Murgatroyd's metabolism was very close to man's. If he ate something and it didn't disagree with him, it was very likely safe for a man to eat it, too. If he rejected something, it probably wasn't. But his real value was much more important than the tasting of questionable foods.

When Calhoun camped the first night, he made a fire of a plant shaped like a cactus-barrel and permeated with oil. By heaping dirt around it, he confined its burning to a round space very much like the direct-heat element of an electronic stove. It was an odd illustration of the fact that human progress does not involve anything really new in kind, but only increased convenience and availability of highly primitive comforts. By the light of

that circular bonfire, Calhoun actually read a little. But the light was inadequate. Presently he yawned. One did not get very far in the Med Service without knowing probability in human conduct. It enabled one to check on the accuracy of statements made, whether by patients or officials, to a Med Ship man. Today, though, he'd traveled a long way on foot. He glanced at Murgatroyd, who was gravely pretending to read from a singularly straight-edged leaf.

"Murgatroyd," said Calhoun, "it is likely that you will interpret any strange sound as a possible undesirable subjective experience. Which is to say, as dangerous. So if you hear anything sizable coming close during the night, I hope you'll squeal. Thank you."

Murgatroyd said "*Chee*," and Calhoun rolled over and went to sleep.

It was mid-morning of the next day when he came upon a cultivated field. It had been cleared and planted, of course, in preparation for the colonists who'd been expected to occupy the city. Familiar Earth-plants grew in it, ten feet high and more. And Calhoun examined it carefully, in the hope of finding how long since it had received attention. In his examination, he found the dead man.

As a corpse, the man was brand-new, and Calhoun very carefully put himself into a strictly medical frame of mind before he bent over for a technical estimate of what had happened, and when. The dead man seemed to have died of hunger. He was terribly emaciated, and he didn't belong in a cultivated field far from the city. By his garments he was a city-dweller and a prosperous one. He wore the jewels which nowadays indicated a man's profession and status in it much more than the value of his possessions. There was money in his pockets, and writing materials, a wallet with pictures and identification, and the normal oddments a man would carry. He'd been a civil servant. And he shouldn't have died of starvation.

He especially shouldn't have gone hungry here! The sweet-maize plants were tall and green. Their ears were ripe. He hadn't gone hungry! There were the inedible remains of at least two dozen sweet-maize ears. They had been eaten some time—some days—ago, and one had been left unfinished. If the

dead man had eaten them but was unable to digest them, his belly should have been swollen with undigested food. It wasn't. He'd eaten and digested and still had died, at least largely of inanition.

Calhoun scowled.

"How about this corn, Murgatroyd?" he demanded.

He reached up and broke off a half-yard-long ear. He stripped away the protecting, stringy leaves. The soft grains underneath looked appetizing. They smelled like good fresh food. Calhoun offered the ear to Murgatroyd.

The little *tormal* took it in his paws and on the instant was eating it with gusto.

"If you keep it down, he didn't die of eating it," said Calhoun, frowning, "and if he ate it—which he did—he didn't die of starvation. Which he did."

He waited. Murgatroyd consumed every grain upon the oversized cob. His furry belly distended a little. Calhoun offered him a second ear. He set to work on that, too, with self-evident enjoyment.

"In all history," said Calhoun, "nobody's ever been able to poison one of you *tormals* because your digestive system has a qualitative-analysis unit in it that yells bloody murder if anything's likely to disagree with you. As a probability of *tormal* reaction, you'd have been nauseated before now if that stuff wasn't good to eat."

But Murgatroyd ate until he was distinctly pot-bellied. He left a few grains on the second ear with obvious regret. He put it down carefully on the ground. He shifted his left-hand whiskers with his paw and elaborately licked them clean. He did the same to the whiskers on the right-hand side of his mouth. He said comfortably:

"*Chee!*"

"Then that's that," Calhoun told him. "This man didn't die of starvation. I'm getting queasy!"

He had his lab kit in his shoulder-pack, of course. It was an absurdly small outfit, with almost microscopic instruments. But in Med Ship field work the techniques of microanalysis were standard. Distastefully, Calhoun took the tiny tissue-sample from which he could gather necessary information. Standing, he ran through the analytic process that seemed called for. When

he finished, he buried the dead man as well as he could and started off in the direction of the city again. He scowled as he walked.

He journeyed for nearly half an hour before he spoke. Murgatroyd accompanied him on all fours, now, because of his heavy meal. After a mile and a half, Calhoun stopped and said grimly:

"Let's check you over, Murgatroyd."

He verified the *tormal's* pulse and respiration and temperature. He put a tiny breath-sample through the part of the lab kit which read off a basic metabolism rate. The small animal was quite accustomed to the process. He submitted blandly. The result of the checkover was that Murgatroyd the *tormal* was perfectly normal.

"But," said Calhoun angrily, "that man died of starvation! There was practically no fat in that tissue-sample at all! He arrived where we found him while he was strong enough to eat, and he stayed where there was good food, and he ate it, and he digested it, and he died of starvation! Why?"

Murgatroyd wriggled unhappily, because Calhoun's tone was accusing. He said, "*Chee!*" in a subdued tone of voice. He looked pleadingly up at Calhoun.

"I'm not angry with you," Calhoun told him, "but dammit—"

He packed the lab kit back into his pack, which contained food for the two of them for about a week.

"Come along!" he said bitterly. He started off. Ten minutes later he stopped. "What I said was impossible. But it happened, so it mustn't have been what I said. I must have stated it wrongly. He could eat, because he did. He did eat, because of the cobs left. He did digest it. So why did he die of starvation? Did he stop eating?"

"*Chee!*" said Murgatroyd, with conviction.

Calhoun grunted and marched on once more. The man had not died of a disease—not directly. The tissue analysis gave a picture of death which denied that it came of any organ ceasing to function. Was it the failure of the

organism—the man—to take the action required for living? Had he stopped eating?

Calhoun's mind skirted the notion warily. It was not plausible. The man had been able to feed himself and had done so. Anything which came upon him and made him unable to feed himself—

"He was a city man," growled Calhoun. "And this is a long way from the city. What was he doing away out here, anyhow?"

He hesitated and tramped on again. A city man found starved in a remote place might have become lost, somehow. But if this man was lost, he was assuredly not without food.

"If there was a ground-car," Calhoun considered, "it wouldn't mean anything. If he dared go back to the city he might have used it, but he wouldn't have been where I found him if he hadn't wanted or needed to leave the city. Hm-m—He walked out into the middle of the field. He was hungry—why didn't he have food?—and he ate. He stayed there for days, judging by the amount of food he ate *and* digested. Why did he do that? Then he stopped eating and died. Again why?"

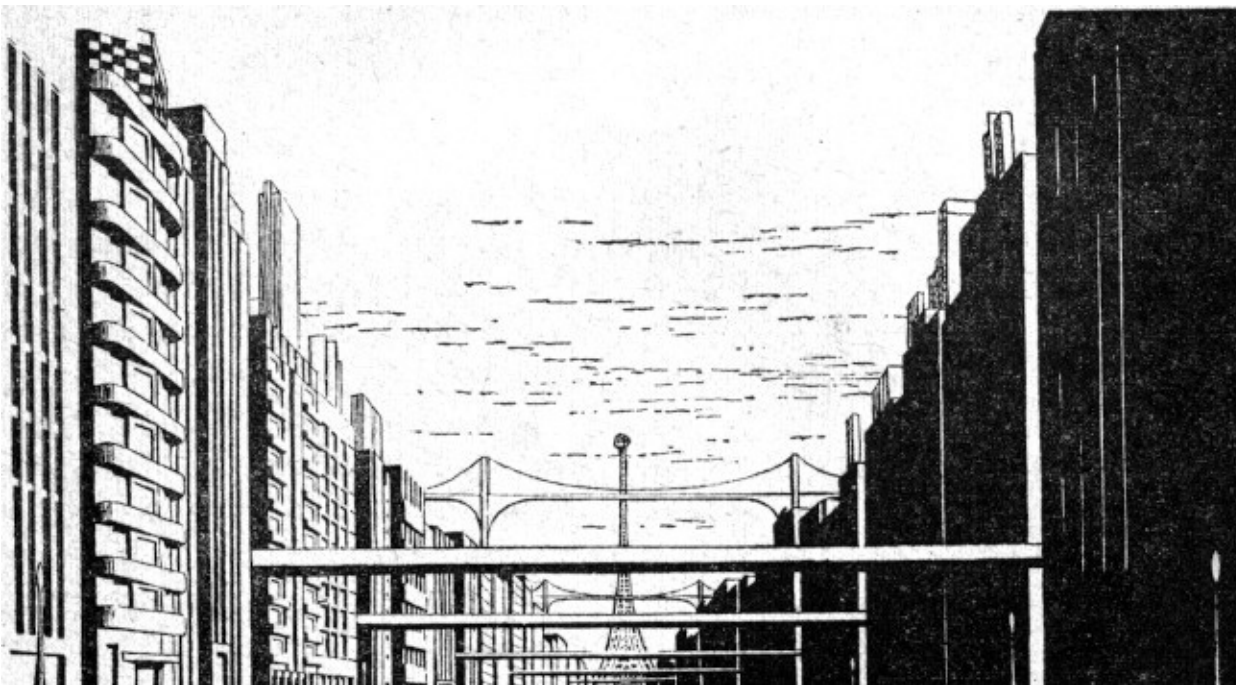
He crossed over the top of a rounded hillock some three miles from the shallow grave he'd made. He began to accept the idea that the dead man had stopped eating, for some reason, as the only possible explanation. But that didn't make it plausible. He saw another ridge of higher hills ahead.

In another hour he came to the crest of that farther range. It was the worn-down remnant of a very ancient mountain-range, now eroded to a mere fifteen hundred or two thousand feet. He stopped at the very top. Here was a time and place to look and take note of what he saw. The ground stretched away in gently rolling fashion for very many miles, and there was the blue blink of sea at the horizon. A little to the left he saw shining white. He grunted.

That was the city of Maris III, which had been built to receive colonists from Dettra and relieve the population-pressure there. It had been planned as the nucleus of a splendid, spacious, civilized world-nation to be added to the number of human-occupied worlds. From its beginning it should hold a population in the hundreds of thousands. It was surrounded by cultivated

fields, and the air above it should be a-shimmer with flying things belonging to its inhabitants.

Calhoun stared at it through his binoculars. They could not make an image, even so near, to compare to that the electron telescope had made from space, but he could see much. The city was perfect. It was intact. It was new. But there was no sign of occupancy anywhere in it. It did not look dead, so much as frozen. There were no fliers above it. There was no motion on the highways. He saw one straight road which ran directly away along his line of sight. Had there been vehicles on it, he would have seen at least shifting patches of color as clots of traffic moved together. There were none.



He pressed his lips together. He began to inspect the nearer terrain. He saw foreshortened areas where square miles of ground had been cleared and planted to Earth vegetation. The ground would have been bulldozed clean, and then great sterilizers would have lumbered back and forth, killing every native seed and root and even the native soil-bacteria. Then there would have been spraying with cultures of the nitrogen-fixing and phosphorous-releasing microscopic organisms which normally lived in symbiosis with Earth-plants. They would have been tested beforehand for their ability to compete with indigenous bacterial life. And then Earth-plants would have been seeded.

They had been. Calhoun saw that inimitable green which a man somehow always recognizes. It is the green of plants whose ancestors thrive on Earth and have followed that old planet's children halfway across the galaxy.

"The population must be practically nothing," growled Calhoun, "because it doesn't show. But the part of it in the city wants to keep whatever's happened from the Med Service. Hm-m-m. They're not dying, or they'd want help. But at least one dead man wasn't in the city where he belonged, and he could have used some help! Maybe there are more like him."

Murgatroyd said,

"Chee!"

"If there are two kinds of people here," added Calhoun darkly, "they might be—antagonistic to each other."

He stared with knitted brows over the vast expanse toward the horizon. Murgatroyd had halted a little behind him. He stood up on his hind legs and stared intently off to one side. He shaded his eyes with a forepaw in a singularly humanlike fashion and looked inquisitively at something he saw. But Calhoun did not notice.

"Make a guess, Murgatroyd," said Calhoun. "There are at least a few people in the city who don't want something known to the Med Service. So whatever's the matter, it's not fatal to them. There may be people wandering about like that poor devil we found. Something was fatal to him! Where'd we find more of his type? Since they haven't tried to kill me, we might make friends."

Murgatroyd did not answer. He stared absorbedly at a patch of underbrush some fifty yards to the left.

Calhoun shrugged and started down the hillside. Murgatroyd remained fixed in a pose of intensely curious attention to the patch of brush. Calhoun went on down the farther hillside. His back was toward the brush-thicket.

There was a deep-toned, musical twanging sound from the thicket. Calhoun's body jerked violently as an impact sounded. He stumbled and went down,

with the shaft of a wooden projectile sticking out of his pack. He lay still.

Murgatroyd whimpered. He rushed to where Calhoun lay upon the ground. He danced in agitation, chattering shrilly. He wrung his paws in humanlike distress. He whimpered and chattered together. He tugged at Calhoun. Calhoun made no response.

A figure came out of the thicket. It was gaunt and thin, yet its garments had once been of admirable quality. It carried a strange and utterly primitive weapon. It moved toward Calhoun without lightness, but with a dreary resolution.



It bent over him and laid a hand to the wooden projectile it had fired into his back.

Calhoun moved suddenly. He grappled. The gaunt figure toppled, and he swarmed upon it savagely as it struggled. But it was taken by surprise. Pantings sounded, and Murgatroyd danced in a fever of anxiety.

Then Calhoun stood up quickly. He stared down at the emaciated figure which had tried to murder him from ambush. That figure panted horribly, now.

"Really," said Calhoun in a professional tone, "as a doctor I'd say that you should be in bed instead of wandering around trying to murder total strangers. When did this trouble begin? I'm going to take your temperature and your pulse. Murgatroyd and I have been hoping to find someone like you. The only other human being I've seen on this planet wasn't able to talk."

He swung his shoulder-pack around and impatiently jerked a sharp-pointed stick out of it. It was the missile, which had been stopped by the pack. He brought out his lab kit. With absolute absorption in the task, he prepared to make a swift check of his would-be murderer's state of health.

It was not good. There was already marked emaciation. The desperately panting young woman's eyes were deep-sunk: hollow. She gasped and gasped. Still gasping, she lapsed into unconsciousness.

"Here," said Calhoun curtly, "you enter the picture, Murgatroyd! This is the sort of thing you're designed to handle!"

He set to work briskly. But presently he said over his shoulder:

"Besides a delicate digestion and a hair-trigger antibody system, Murgatroyd, you ought to have the instincts of a watchdog. I don't like coming that close to being speared by my patients. See if there's anybody else around, won't you?"

"*Chee!*" said Murgatroyd shrilly. But he didn't understand. He watched as Calhoun deftly drew a small sample of blood from the unconscious young woman and painstakingly put half the tiny quantity into an almost microscopic ampule in the lab kit. Then he moved toward Murgatroyd.

The *tormal* wriggled as Calhoun made the injection. But it did not hurt. There was an insensitive spot on his flank where the pain-nerves had been blocked off before he was a week old.

"As one medical man to another," said Calhoun, "what's a good treatment for anoxia when you haven't got any oxygen? You don't know? Neither do I. But we've found out why those chaps in the city tried to shake us to bits, out in space."

He swore in a sudden, bitter anger. Then he looked quickly at the girl, concerned lest she'd heard.

She hadn't. She was still unconscious.

III

"That pattern of human conduct which is loosely called "self-respecting" has the curious property of restricting to the individual—through his withdrawal of acts to communicate misfortune—the unfavorable chance occurrences which probability insists must take place. On the other hand, the same pattern of human conduct tends to disseminate and to share chance favorable occurrences among the group. The members of a group of persons practicing "self-respect," then, increase the mathematical probability of good fortune to all their number. This explains the instability of cultures in which principles leading to this type of behavior become obsolete. A decadent society brings bad luck upon itself by the operation of the laws of probability...."

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She came very slowly back to consciousness. It was almost as if she waked from utterly exhausted sleep. When she first opened her eyes, they wandered vaguely until they fell upon Calhoun. Then a bitter and contemptuous hatred filled them. Her hand fumbled weakly to the knife at her waist. It was not a good weapon. It had been table-cutlery and the handle was much too slender to permit a grip by which somebody could be killed. Calhoun bent over and took the knife away from her. It had been ground unskillfully to a point.

"In my capacity as your doctor," he told her, "I must forbid you to stab me. It wouldn't be good for you." Then he said, "Look! My name's Calhoun. I came from Sector Med Headquarters to make a planetary health inspection, and some lads in the city apparently didn't want a Med Ship aground. So they tried to kill me by buttering me all over the walls of my ship, with the landing-grid field. I made what was practically a crash landing, and now I need to know what's up."

The burning hatred remained in her eyes, but there was a trace of doubt.

"Here," said Calhoun, "is my identification."

He showed her the highly official documents which gave him vast authority—where a planetary government was willing to concede it.

"Of course," he added, "papers can be stolen. But I have a witness that I'm what and who I say I am. You've heard of *tormals*? Murgatroyd will vouch for me."

He called his small and furry companion. Murgatroyd advanced and politely offered a small, prehensile paw. He said "*Chee*" in his shrill voice, and then solemnly took hold of the girl's wrist in imitation of Calhoun's previous action of feeling her pulse.

Calhoun watched. The girl stared at Murgatroyd. But all the galaxy had heard of *tormals*. They'd been found on a planet in the Deneb region, and they were engaging pets and displayed an extraordinary immunity to the diseases men were apt to scatter in their interstellar journeyings. A forgotten Med Service researcher made an investigation on the ability of *tormals* to live in contact with men. He came up with a discovery which made them very much too valuable to have their lives wasted in mere sociability. There were still not enough of Murgatroyd's kind to meet the need that men had of them, and laymen had to forego their distinctly charming society. So Murgatroyd was an identification.

The girl said faintly:

"If you'd only come earlier.... But it's too late now! I ... thought you came from the city."

"I was headed there," said Calhoun.

"They'll kill you—"

"Yes," agreed Calhoun, "they probably will. But right now you're ill and I'm Med Service. I suspect there's been an epidemic of some disease here, and that for some reason the people in the city don't want the Med Service to know about it. You seem to have ... whatever it is. Also you had a very curious weapon to shoot me with."

The girl said drearily:

"One of our group had made a hobby of such things. Ancient weapons. He had bows and arrows and—what I shot you with was a crossbow. It doesn't

need power. Not even chemical explosives. So, when we ran away from the city, he ventured back in and armed us as well as he could."

Calhoun nodded. A little irrelevant talk is always useful at the beginning of a patient-interview. But what she said was not irrelevant. A group of people had fled the city. They'd needed arms, and one of their number had "ventured" back into the city for them. He'd known where to find only reconstructions of ancient lethal devices—a hobby collection. It sounded like people of the civil-service type. Of course there were no longer social classes separated by income. Not on most worlds, anyhow. But there were social groupings based on similar tastes, which had led to similar occupations and went on to natural congeniality. Calhoun placed her, now. He remembered a long-out-moded term, "upper middle class" which no longer meant anything in economics but did in medicine.

"I'd like a case-history," he said conversationally. "Name?"

"Helen Jons," she said wearily.

He held the mike of his pocket recorder to pick up her answers. Occupation, statistician. She'd been a member of the office force which was needed during the building of the city. When the construction work was finished, most of the workmen returned to the mother-world Dettra, but the office staff stayed on to organize things when colonists should arrive.

The plague appeared among the last shipload of workmen waiting to be returned to the mother world. There were about a thousand persons in the city altogether. The disease produced, at first, no obvious physical symptoms, but those afflicted with it tended to be listless and lackadaisical and without energy. The first-noticed symptom was a cessation of gripes and quarrelings among the workmen. Shortness of breath appeared two days later. It was progressive. Deaths began in two weeks. Men sank into unconsciousness and died. By the time the transport-ship arrived from Dettra with colonists to be landed ... it was to take back the workmen ... the physicians on the planet were grim. They described the situation by space phone. The transport returned to Dettra without removing the workmen or

landing the colonists. The people left in the city on Maris III were self-quarantined, but they expected help.

It was two months before another ship arrived. By then fewer than two hundred of the original thousand remained. More than half those survivors were already listless and short-breathed. A good ten per cent were in the beginning of that marked lethargy which deepened into coma and ended fatally. A desperate, gaunt, plague-stricken few still manned the landing grid.

The ship came down. Men disembarked. There was no crowd to greet them. The survivors still in the city had scattered themselves widely, hoping to escape the contagion by isolating themselves in new and uncontaminated dwelling-units. But there was no lack of communication facilities. Nearly all the survivors watched on vision screens in contact with the landing grid.

The newcomers did not look like doctors, nor act like them. Visiphone contact with the landing grid was immediately broken. It could not be restored. So the isolated groups spoke agitatedly to each other by other visiphone contacts, exchanging messages of desperate hope. Then, new-landed men appeared at an apartment whose occupant was in the act of such a conversation with a group in a distant building. He left the visiphone on as he went to admit and greet the men he hoped were researchers, at least, come to find the cause of the plague and end it.

The viewer at the other visiphone plate gazed eagerly into his friend's apartment. He saw a group of the newcomers admitted. He saw them deliberately murder his friend and the survivors of his family.

Plague-stricken or merely terrified people—in pairs or trios widely separated through the city—communicated in swift desperation. It was possible that there had been a mistake—a blunder; an unauthorized crime had been committed. But it was not a mistake. Unthinkable as such an idea was, there developed evidence that the plague on Maris III was to be ended as if it were an epizootic among animals. Those who had it and those who had been exposed to it were to be killed to prevent its spread among the newcomers.

A conviction of such horror could not be accepted without absolute proof. But when night fell, the public power-supply of the city was cut off—communications ended. The singular sunset hush of Maris III left utter stillness everywhere—and there were screams which echoed among the city's innumerable empty-eyed, unoccupied buildings.

The scant remainder of the plague-survivors fled in the night. They fled singly, carrying the plague with them. Some carried members of their families already stricken. Some helped already-doomed wives or friends or husbands to the open country. Flight would not save their lives. It would only prevent their murder. But somehow that seemed a thing to be attempted.

"This," said Calhoun, "is not a history of your own case. When did you develop the disease ... whatever it may be?"

"Don't you know what it is?" asked Helen hopelessly.

"Not yet," admitted Calhoun. "I've very little information. I'm trying to get more now."

What other information he had he'd gathered from a newly dead man in a field some miles away. He did not mention that at this moment.

The girl went on, exhaustedly. The first symptom was listlessness, of which the victim was unconscious. One could pull out of it with an effort, but one wasn't aware that anything was wrong. The listlessness progressed. One could realize it only by recognizing the more urgent, more violent effort needed to pay attention, and the discovery of weakness when one tried to act. One did not feel discomfort—not even hunger or thirst. One had to summon increasing resolution even to become aware of the need to do anything at all.

The symptoms were singularly like those of a man too long at too high an altitude without oxygen. They were even more like those of a man in a non-pressurized flier whose oxygen supply has been cut off. But such a man would pass out without realizing that he was slipping into unconsciousness. On Maris III the process was infinitely gradual. It was a matter of two weeks or more.

"I'd been infected before we ran away," she said drearily. "I didn't know it then. Now I know I've a few more days of being able to think and act ... if I try hard enough. But it'll be less and less each day. Then I'll stop being able to try."

Calhoun watched the tiny recorder roll its multiple-channel tape from one spool to the other as she talked.

"You had energy enough to try to kill me," he observed.

He looked at the weapon. There was an arched steel spring placed crosswise at the end of a barrel like a sporting blast-rifle. Now he saw a handle and a ratchet by which the spring was brought to tension, storing up power to throw the missile. He asked:

"Who wound up this crossbow?"

Helen hesitated.

"Kim ... Kim Walpole."

"You're not a solitary refugee now? There are others of your group still alive?"

She hesitated again, and then said:

"Some of us came to realize that staying apart didn't matter. We ... couldn't hope to live, anyhow. We ... already had the plague. Kim is ... one of us. He's the strongest. He ... wound up the crossbow for me. He ... had the weapons to begin with."

Calhoun asked seemingly casual questions. She told him of a group of fugitives remaining together because all were already doomed. There had been eleven of them. Two were dead, now. Three others were in the last lethargy. It was impossible to feed them. They were dying. The strongest was Kim Walpole, who'd ventured back into the city to bring out weapons for the rest. He'd led them, and now was still the strongest and—so the girl considered—the wisest of them all.

They were waiting to die. But the newcomers to the planet—the invaders, they believed—were not content to let them wait. Groups and single hunters came out of the city and searched for them.

"Probably," said the girl dispassionately, "to burn our bodies against contagion. They ... kill us so they won't have to wait. And it's just ... seemed so horrible that we ... felt we ought to defend our right to die naturally by ... dying fighting. That's why I ... shot at you. I shouldn't have, but—"

She stopped, helplessly. Calhoun nodded.

The fugitives now aided each other simply to avoid murder. They gathered together exhaustedly at nightfall, and those who were strongest did what they could for the others. By day, those who could walk scattered to separate hiding places, so that if one were discovered, the others might still escape the indignity of being butchered. They had no stronger motive than that. They were merely trying to die with dignity, instead of being killed as sick beasts. Which bespoke a tradition and an attitude which Calhoun approved. People like these would know something of the science of probability in human conduct. Only they would call it ethics. But the strangers—the invaders—the occupiers of the city were of another type. They probably came from another world.

"I don't like this," said Calhoun coldly. "Just a moment."

He went over to Murgatroyd. Murgatroyd seemed to droop a little. Calhoun checked his breathing and listened to his heart. Murgatroyd submitted, saying only "*Chee*" when Calhoun put him down.

"I'm going to help you to your rendezvous," said Calhoun abruptly. "Murgatroyd's got the plague now. I ... exposed him to it, and he's reacting fast. And I want to see the others of your group before nightfall."

The girl just managed to get to her feet. Even speaking had tired her, but she gamely though wearily moved off at a slant to the hillside's slope. Calhoun picked up the odd weapon and examined it thoughtfully. He wound it up as it was obviously meant to be. He picked up the missile it had fired, and put it in place. He went after the girl, carrying it. Murgatroyd brought up the rear.

Within a quarter of a mile the girl stopped and clung swaying to the trunk of a slender tree. It was plain that she had to rest, and dreaded getting off her feet because of the desperate effort needed to arise.

"I'm going to carry you," said Calhoun firmly. "You tell me the way."

He picked her up bodily and marched on. She was light. She was not a large girl, but she should have weighed more. Calhoun still carried the quaint ancient-type weapon without difficulty.

Murgatroyd followed as Calhoun went up a small inclination on the greater hillside and down a very narrow ravine. Through brushwood he pushed until he came to a small open space where shelters had been made for a dozen or so human beings. They were utterly primitive—merely roofs of leafy

branches over frameworks of sticks. But of course they were not intended for permanent use. They were meant only to protect plague-stricken folk while they waited to die.

But there was disaster here. Calhoun saw it before the girl could. There were beds of leaves underneath the shelters. There were three bodies lying upon them. They would be those refugees in the terminal coma which—since the girl had described it—accounted for the dead man Calhoun had found, dead of starvation with food-plants all around him. But now Calhoun saw something more. He swung the girl swiftly in his arms so that she would not see. He put her gently down and said:

"Stay still. Don't move. Don't turn."

He went to make sure. A moment later he raged. Because it was Calhoun's profession to combat death and illness in all its forms. He took his profession seriously. And there are defeats, of course, which a medical man has to accept, though unwillingly. But nobody in the profession, and least of all a Med Ship man, could fail to be roused to fury by the sight of people who should have been his patients, lying utterly still with their throats cut.

He covered them with branches. He went back to Helen.

"This place has been found by somebody from the city," he told her harshly. "The men in coma have been murdered. I advise you not to look. At a guess, whoever did it is now trying to track down the rest of you."

He went grimly to the small open glade, searching the ground for footprints. There was ground-cover at most places, but at the edge of the clearing he found one set of heavy footprints going away. He put his own foot beside a print and rested his weight on it. His foot made a lesser depression. The other print had been made by a man weighing more than Calhoun. Therefore it was not one of the party of plague-victims.

He found another set of such footprints, entering the glade from another spot.

"One man only," he said icily. "He won't think he has to be on guard, because a city's administrative personnel—such as were left behind for the

plague to hit—doesn't usually have weapons among their possessions. And he's confident that all of you are weak enough not to be dangerous to him."

Helen did not turn pale. She was pale before. She stared numbly at Calhoun. He looked grimly at the sky.

"It'll be sunset within the hour," he said savagely. "If it's the intention of the newcomers ... the invaders to burn the bodies of all plague victims, he'll come back here to dispose of these three. He didn't do it before lest the smoke warn the rest of you. But he knows the shelters held more than three people. He'll be back!"

Murgatroyd said "*Chee!*" in a bewildered fashion. He was on all fours, and he regarded his paws as if they did not belong to him. He panted.

Calhoun checked him over. Respiration way up. Heart-action like that of the girl Helen. His temperature was not up, but down. Calhoun said remorsefully:

"You and I, Murgatroyd, have a bad time of it in our profession. But mine is the worse. You don't have to play dirty tricks on me, and I've had to, on you!"

Murgatroyd said "*Chee!*" and whimpered. Calhoun laid him gently on a bed of leaves which was not occupied by a murdered man.

"Lie still!" he commanded. "Exercise is bad for you!"

He walked away. Murgatroyd whined faintly, but lay still as if exhausted.

"I'm going to move you," Calhoun told the girl, "so you won't be sighted if that man from the city comes back. And I've got to keep out of sight for a while or your friends will mistake me for him. I count on you to vouch for me later. Basically, I'm making an ambush." Then he explained irritably, "I daren't try to trail him because he might not backtrack to return here!"

He lifted the girl and placed her where she could see the glade in its entirety, but would not be visible. He settled down himself a little distance away. He was acutely dissatisfied with the measures he was forced to take. He could not follow the murderer and leave Helen and Murgatroyd unprotected, even though the murderer might find another victim because he was not trailed. In any case Murgatroyd's life, just now, was more important than the life of any human being on Maris III. On him depended everything.

But Calhoun was not pleased with himself.

There was silence except for the normal noises of living wild things. There were fluting sounds, which later Calhoun would be told, from crawling creatures not too much unlike the land-turtles of Earth. There were deep-bass hummings, which came from the throats of miniature creatures which might roughly be described as birds. There were chirpings which were the cries of what might be approximately described as wild pigs—except that they weren't. But the sun Maris sank low toward the nearer hill-crests, and behind them, and there came a strange, expectant hush over all the landscape. At sundown on Maris III there is a singular period when the creatures of the day are silent and those of the night are not yet active. Nothing moved. Nothing stirred. Even the improbable foliage was still.

It was into this stillness and this half-light that small and intermittent rustling sounds entered. Presently there was a faint murmur of speech. A tall, gaunt young man came out of the brushwood, supporting a pathetically feeble old man, barely able to walk. Calhoun made a gesture of warning as the girl Helen opened her lips to speak. The slowly moving pair—the young man moving exhaustedly, the older man staggering with weakness despite his help—came into the glade. The younger helped the older to sit down. He stood panting.

A woman and a man came together, assisting each other. There was barely light enough from the sun's afterglow to show their faces, emaciated and white.

A fifth feeble figure came tottering out of another opening in the brush. He was dark-bearded and broad, and he had been a powerful man. But now the plague lay heavily upon him.

They greeted each other listlessly. They had not yet discovered those of their number who had been murdered.

The gaunt young man summoned his strength and moved toward the shelter where Calhoun had covered an unseemly sight with branches.

Murgatroyd whimpered.

There came another rustling sound. But this had nothing of feebleness in it. Someone pushed branches forthrightly out of his way. He came striding confidently into the small open space. He was well-fleshed, and his color was excellent. Calhoun automatically judged him to be in superlative good health, slightly over-fleshed, and of that physical type which suffers very few psychosomatic troubles because it lives strictly and enjoyably in the present.

Calhoun stood up. He stepped out into the fading light just as the sturdy last-comer grinned at the group of plague-stricken semi-skeletons.

"Back, eh?" he said amiably. "Saved me a lot of trouble. I'll make one job of it."

With leisurely confidence he reached to the blaster at his hip.

"Drop it!" snapped Calhoun, from quartering behind him. "Drop it!"

The sturdy man whirled. He saw Calhoun with a crossbow raised to cover him. There was light enough to show that it was not a blast-rifle—in fact, that it was no weapon of any kind modern men would ordinarily know. But much more significant to the sturdy man was the fact that Calhoun wore a uniform and was in good health.

He snatched out his blast-pistol with professional alertness.

And Calhoun shot him with the crossbow. It happened that he shot him dead.

IV

"Statistically, it must be recognized that no human action is without consequences to the man who acts. Again statistically, it must be recognized that the consequences of an action tend with strong probability to follow the general pattern of the action. A violent action, for example, has a strong probability of violent consequences, and since some at least of the consequences of an act must affect the person acting, a man who acts violently exposes himself to the probability that chance consequences which affect him, if unfavorable, will be violently so."

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Murgatroyd had been inoculated with a blood-sample from the girl Helen some three hours or less before sunset. But it was one of the more valuable genetic qualities of the *tormal* race that they reacted to bacterial infection as a human being reacts to medication. Medicine on the skin of a human being rarely has any systemic effect. Medication on mucus membrane penetrates better. Ingested medication—medicine that is swallowed—has greater effectiveness still. But substances injected into tissues or the blood-stream have most effect of all. A centigram of almost any drug administered by injection will have an effect close to that of a gram taken orally. It acts at once and there is no modification by gastric juices.

Murgatroyd had had half a cubic centimeter of the girl's blood injected into the spot on his flank where he could feel no pain. It contained the unknown cause of the plague on Maris III. Its effect as injected was incomparably greater than the same infectious material smeared on his skin or swallowed. In either such case, of course, it would have had no effect at all, because *tormals* were to all intents and purposes immune to ordinary contagions. Just as they had a built-in unit in their digestive tract to cause the instant rejection of unwholesome food, their body-cells had a built-in ability to produce antibodies immediately the toxin of a pathogenic organism came into contact with them. So *tormals* were effectively safe against any disease transmitted by ordinary methods of infection. Yet if a culture of pathogenic bacteria—say—were injected into their blood stream, their whole body set to work to turn out antibodies because all their body was attacked. And all at once. There was practically no incubation-period.

Murgatroyd, who had been given the plague in mid-afternoon, was reacting violently to its toxins by sunset. But two hours after darkness fell he arose and said shrilly, "*Chee-chee-chee!*" He'd been sunk in heavy slumber. When he woke, there was a small fire in the glade, about which the exhausted, emaciated fugitives consulted with Calhoun. Calhoun was saying bitterly:

"Those characters in the city are immune! They have to be! And they know they're immune, or they wouldn't risk contagion by murdering you or handling the bodies of plague-victims to burn them! So they have to know all about the plague—and they knew it before they arrived! They came because they knew! That's why I shot that man with the crossbow, instead of

taking a blaster to him. I meant to wound him so I could make him answer questions, but it's not an accurate weapon and I killed him instead. I got very little from the stuff in his pockets. The only significant thing was a ground-car key, and even that means only there's a car waiting somewhere for him."

The gaunt young man said drearily:

"He didn't come from Dettra, our home planet. Fashions are different on different worlds. His foot-wear was like a style we had on Dettra four years back, and his body-clothing has fasteners we don't use."

Murgatroyd saw Calhoun and rushed to him, embracing his legs with enthusiasm and chattering shrilly of his relief at finding the man he knew. The skeletonlike plague-victims stared at him.

"This," said Calhoun with infinite relief, "is Murgatroyd. He's had the plague and is over it. So now we'll get you people cured. I wish I had better light!"

He counted Murgatroyd's breathing and listened to his heart. Murgatroyd was in that state of boisterous good health which is normal in any lower animal, but amounts to genius in a *tormal*. Calhoun regarded him with satisfaction.

"All right!" he said. "Come along!"

He plucked a brand of burning resinous stuff from the campfire. He handed it to the gaunt young man and led the way. Murgatroyd ambled complacently after him. Calhoun stopped under one of the unoccupied shelters and got out his lab kit. He bent over Murgatroyd. What he did, did not hurt. When he stood up, he squinted at the red fluid in the instrument he'd used.

"About fifteen CCs," he observed. "This is strictly emergency stuff I'm doing now. But I'd say that there's an emergency."

The gaunt young man said:

"I'd say you've doomed yourself. The incubation-period seems to be about six days. It took that long to develop among the doctors we had in the office staff."

Calhoun opened a compartment of the kit, whose minuscule test tubes and pipettes gleamed in the torchlight. He absordedly transferred the reddish fluid to a miniature filter-barrel, piercing a self-healing plastic cover to do so. He said:

"You're pre-med? The way you talk—"

"I was an interne," said Kim. "Now I'm pre-corpse."

"I doubt that last," said Calhoun. "But I wish I had some distilled water—This is anticoagulant." He added the trace of a drop to the sealed, ruddy fluid. He shook the whole filter to agitate it. The instrument was hardly larger than his thumb. "Now a clumper—" He added a minute quantity of a second substance from an almost microscopic ampule. He shook the filter again. "You can guess what I'm doing. With a decent lab I'd get the structure and formula of the antibody Murgatroyd has so obligingly turned out for us. We'd set to work to synthesize it. In twenty hours, lab time, we ought to have it coming out of the reaction-flasks in quantity. But there is no lab."

"There's one in the city," said the gaunt young man hopelessly. "It was for the colonists who were to come. And we were staffed to give them proper medical care. When the plague came, our doctors did everything imaginable. They not only tried the usual culture tricks, but they cultured samples of every separate tissue in the fatal cases. They never found a single organism—even in the electron microscopes—that would produce the plague." He said with a sort of weary pride, "Those who'd been exposed worked until they had it, then others took over. Every man worked as long as he could make his brain work, though."

Calhoun squinted through the glass tube of the filter at the sputtering torch.

"Almost clumped," he observed. Then he said, "Did you ever hear of a man named Pasteur? One of his first discoveries was that one could get an effectively pure culture of a pathogenic organism by giving the disease to an experimental animal. Better ways were found later, but one still expects a pure culture in a patient who has a disease really badly. What did the lab turn up?"

Kim shook his head.

"Nothing. The bacteriological survey of the planet had been thorough. Oral and intestinal flora were normal. Naturally, the local bacteria couldn't

compete with the strains we humans have learned to live with. They couldn't symbiotize. So there wasn't anything unknown. There wasn't any cause of the plague."

Calhoun began to work the filter plunger, by the wavering light of the torch. The piston was itself the filter, and on one side a clear, mobile liquid began very slowly to appear.

"Mutated standard bug? Still, if your doctors did cultures and couldn't reproduce the disease—"

"They could pass it," said Kim bitterly, "but they could never find what carried it! No pure culture would!"

Calhoun watched the clear fluid develop on the delivery side of the filtering piston. The job got done. There was better than twelve cubic centimeters of clear serum on the delivery side, and an almost solid block of clumped blood cells on the other. He drew off the transparent fluid with a fine precision.

"We're doing biochemistry under far from aseptic conditions," he said wryly, "but the work has to be done and we have to take the risk. Anyhow, I'm getting a feeling that this isn't any ordinary plague. A normal pathogenic organism should have been turned up by your doctors."

"It wasn't," said Kim.

"So," said Calhoun, "maybe it isn't one isolatable organism. Maybe the disease-producing mechanism simply isn't there when you make pure cultures of the separate strains of virus and microbe. Murgatroyd was a pretty sick animal. I've only known of one previous case in which a *tormal* reacted as violently as Murgatroyd did. That one had us sweating."

"If I were going to live," said Kim grimly, "I might ask what it was."

"Since you're going to," Calhoun told him, "I'll tell you though you don't. It was a pair of organisms. Their toxins acted synergically together. Separately they were innocuous. Together they were practically explosive. That one was the devil to track down!"

He went back across the glade. Murgatroyd came skipping after him, scratching at the anaesthetic patch on his hide, which he sometimes seemed to notice not because it felt oddly, but because it did not feel at all.

"You," said Calhoun briefly to Helen Jons, "you go first. This is an antibody serum. You may itch afterward, but I doubt it. Your arm, please."

She bared her rather pitifully thin arm. He gave her practically a CC of fluid which—plus blood-corpuscles and some forty-odd other essential substances—had been circulating in Murgatroyd's blood stream not long since. The blood-corpuscles had been clumped and removed by one compound plus the filter, and the anticoagulant had neatly modified most of the others. In a matter of minutes, the lab kit had prepared as usable a serum as any animal-using technique would produce. Logically, the antibodies it contained should be isolated and their chemical structure determined. They should be synthesized, and the synthetic antibody-complex administered to plague-victims. But Calhoun faced a small group of people doomed to die. He could only use his field kit to produce a small-scale miracle for them. He could not do a mass production job.

"Next!" said Calhoun. "Tell them what it's all about, Kim!"

The gaunt young man bared his own arm.

"If what he says is so, this will cure us. If it isn't so, nothing can do us any harm!"

And Calhoun briskly gave them, one after another, the shots of what ought to be a curative serum for an unidentified disease which he suspected was not caused by any single germ, but by a partnership. Synergy is an acting-together. Charcoal will burn quietly. Liquid air will not burn at all. But the two together constitute a violent explosive. This is analogous to synergy. The ancient simple drug sulfa is not intoxicating. A glass of wine is not intoxicating. But the two together have the kick of dynamite. Synergy, in medicine, is a process in which when one substance with one effect is given in combination with another substance with another effect, the two together have the consequences of a third substance intensified to fourth or fifth or tenth power.

"I think," said Calhoun when he'd finished, "that by morning you'll feel better—perhaps cured of the plague and only weak from failure to force yourselves to take nourishment. If it turns out that way, I advise you all to get as far away from the city as possible for a considerable while. I think this planet is going to be repopulated. I suspect that shiploads of colonists are on the way here now—but not from Dettra, which built the city. And I definitely guess that, sick or well, you're going to be in trouble if or when you contact the new colonists."

They looked tiredly at him. They were a singular lot of people. Each one seemed half-starved, yet their eyes had not the brightness of suffering. They looked weary beyond belief, and yet there was not self-neglect. They were of that singular human type which maintains human civilization against the inertia of the race—because it drives itself to get needed things done. It is not glamorous, this dogged part of mankind which keeps things going. It is sometimes absurd. For dying folk to wash themselves, when even such exertion calls for enormous resolution, is not exactly rational. To help each other try to die with dignity was much more a matter of self-respect than of intellectual decision. But as a Med Ship man, Calhoun viewed them with some warmth. They were the type that has to be called on when an emergency occurs and the wealth-gathering type tends to flee and the low-time-sense part of a population inclines to riot or loot or worse.

Now they waited listlessly for their own deaths.

"There's no exact precedent for what's happened here," explained Calhoun. "A thousand years or so ago there was a king of France—a country back on old Earth—who tried to wipe out a disease called leprosy by executing all the people who had it. Lepers were a nuisance. They couldn't work. They had to be fed by charity. They died in inconvenient places and only other lepers dared handle their bodies. They tended to throw normal human life out of kilter. That wasn't the case here. The man I killed wanted you dead for another reason. He wanted you dead right away."

The gaunt Kim Walpole said tiredly:

"He wanted to dispose of our bodies in a sanitary fashion."

"Nonsense!" snapped Calhoun. "The city's infected. You lived, ate, breathed, walked in it. Nobody can dare use that city unless they know how the

contagion's transmitted, and how to counteract it. Your own colonists turned back. These men wouldn't have landed if they hadn't known they were safe!"

There was silence.

"If the plague is an intended crime," added Calhoun, "you are the witnesses to it. You've got to be gotten rid of before colonists from somewhere other than Dettra arrive here."

The dark-bearded man growled:

"Monstrous! Monstrous!"

"Agreed," said Calhoun. "But there's no interstellar government, now, any more than there was a planetary government in the old days back on Earth. So if somebody pirates a colony ready to be occupied, there's no authority able to throw them out. The only recourse would be war. And nobody is going to start an interplanetary war! Not with the bombs that can be landed! If the invaders can land a population here, they can keep it here. It's piracy, with nobody able to do anything to the pirates." He paused, and said with irony: "Of course they could be persuaded that they were wrong."

But that was not even worth thinking about. In the computation of probabilities in human conduct, self-interest is a high-value factor. Children and barbarians have clear ideas of justice due to them, but no idea at all of justice due from them. And though human colonies spread toward the galaxy's rim, there was still a large part of every population which was civilized only in that it could use tools. Most people still remained comfortably barbaric or childish in their emotional lives. It was a fact that had to be considered in Calhoun's profession. It bore remarkably on matters of health.

"So you'll have to hide. I think permanently," Calhoun told them. "But in the morning, after I've checked on you people again, I think I'll go into the city and see what I can do about it. Try to rest now. You should all feel much better in the morning."

Kim Walpole said abruptly:

"You've been exposed to the plague. Have you protected yourself?"

"Not yet," acknowledged Calhoun. "Give me a quarter of a CC."

He handed the injector to the gaunt young man. He noted the precision with which Kim handled it. Then he helped get the survivors of the original group—there were six of them—to the leafy beds under the shelters. They were very quiet—even more quiet than their illness demanded. They were very polite. The old man and woman who had struggled back to the glade together made an especial attempt to bid Calhoun good night with the courtesy appropriate to city folk of tradition.

Calhoun settled down to keep watch through the night. Murgatroyd snuggled confidingly close to him. There was silence.

But not complete silence. The night of Maris III was filled with tiny noises, and some not so tiny. There were little squeaks which seemed to come from all directions, including overhead. There were chirpings which were definitely at ground-level. There was a sound like effortful grunting somewhere in the direction of the rampart of hills. In the lowlands there was a rumbling which moved very slowly from one place to another. By its rate of motion, Calhoun guessed that a pack or herd of small animals was making a night-journey and uttering deep-bass noises as it traveled.

He debated certain grim possibilities. The man he'd killed had had a ground-car key in his pocket. He'd probably come out in a powered vehicle. He might have had a companion, and the method of hunting down fugitives—successful, in his case—was probably well established. That companion might come looking for him, so watchfulness was necessary.

Meanwhile—the plague. The idea of synergy was still most plausible. Suppose the toxins—the poisonous metabolic products—of two separate kinds of bacteria combined to lessen the ability of the blood to carry oxygen and scavenge away carbon dioxide? It would be extremely difficult to identify the pair, and the symptoms would be accounted for. No pure culture of any organism to be found would give the plague. Each, by itself, would be harmless. Only a combination of two would be injurious. And if so much was assumed—why—if the blood lost its capacity to carry oxygen, mental listlessness would be the first symptom of all. The brain requires a high oxygen-level in its blood-supply if it is to work properly. Let a man's brain be gradually, slowly, starved of oxygen and all the noted effects would

follow. His other organs would slow down, but at a lesser rate. He would not remember to eat. His blood would still digest food and burn away its own fat—though more and more sluggishly—while his brain worked only foggily. He would become only semiconscious, and then there would come a time of coma when unconsciousness claimed him and his body lived on only as an idling machine—until it ran out of fuel and died.

Calhoun tried urgently to figure out a synergic combination which might make a man's blood cease to do its work. Perhaps only minute quantities of the dual poison might be needed—like an antivitamin or an antienzyme, or —

The invaders of the city were immune. Quite possibly the same antibodies Murgatroyd had produced were responsible for their safety. Somewhere, somebody had very horribly used the science of medicine to commit a monstrous crime. But the science of medicine—

A savage idea came to Calhoun. Its practicality might depend on the number of men in the city. But his eyes burned.

He heard a movement across the glade. He reached for his blaster. Then he saw where the motion was. It was Kim Walpole, intolerably weary, trudging with infinite effort to where Helen Jons lay. Calhoun heard him ask heavily:

"You're all right?"

"Yes, Kim," said the girl softly. "I couldn't sleep. I'm ... wondering if we can hope."

Kim did not answer.

"If we live—" said the girl yearningly, and stopped.

Calhoun felt that he ought to put his fingers in his ears. The conversation was strictly private. But he needed to be on guard. So he coughed, to give notice that he heard. Kim called to him across the starlit glade.

"Calhoun."

"Yes," said Calhoun. "If you two talk, I suggest that you do it in whispers. I want to listen, in case the man I killed had friends who'll come looking for him. Did you get his blaster, by the way?"

"Yes," said Kim from the darkness across the way.

"Good!" said Calhoun. "Keep it. And against all medical ethics, I advise you to use it freely if you find suitable targets. But now, just talk quietly if you can."

He settled back. Murgatroyd stirred and cuddled closer against him without wakening. There was the faintest possible murmuring of voices where Kim Walpole and the girl Helen talked wistfully of the possibility of hope.

Calhoun felt very lonely, despite the violent activities he foresaw for the morrow. He almost envied Kim Walpole. But he could not have traded places with him. It wouldn't have been a fair trade. Calhoun was quite confident that—via Murgatroyd—the folk in the glade had a very fair chance of living for some time yet.

His own chances, considering what he had to do, were more nearly zero. Just about zero, when considered dispassionately.

V

"Very much of physical science is merely the comprehension of long-observed facts. In human conduct, there is a long tradition of observation, but a very brief record of comprehension. For example, human life in contact with other human lives follows the rules of other ecological systems. All too often, however, a man may imagine that an ecological system is composed only of things, whereas such a system operates through the actions of things. It is not possible for any part of an ecological complex to act upon the other parts without being acted upon, in its turn. So that it is singularly stupid—and singularly common—for an individual to consider human society as passive and unreactive, so that he may do what he pleases without a reaction as energetic as his action, and as well-directed. Moreover, probability—."

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An hour after sunrise Calhoun's shoulder-pack was empty of food. The refugees arose, and they were weak and ravenous. Their respiration had slowed to normal. Their pulses no longer pounded. Their eyes were no longer dull, but very bright. But they were in advanced states of malnutrition, and only now were aware of it. Their brains were again receiving adequate oxygen and their metabolism was at a normal level—and they knew that they were starving.

Calhoun served as cook. He trudged to the spring that Helen described and brought back water. While they sucked on sweet tablets from his rations and watched with hungry eyes, he made soup from the dehydrated rations he'd carried for Murgatroyd and himself. He gave it to them as the first thing their stomachs were likely to digest.

He watched as they fed themselves. The elderly man and woman consumed it delicately, looking at each other. The man with the broad dark beard ate with enormous self-restraint. Helen fed the weakest oldest man, between spoonfuls for herself, and Kim Walpole ate slowly, brooding.

Calhoun drew him aside.

"During the night," he said, "I got another lot of serum ready. I'm leaving it with you, with an injector. You'll find other refugees. I gave you massive doses. You'd better be stingy. Try half-CC shots."

"What about you?" demanded Kim.

Calhoun shrugged.

"You'd be surprised how much authority I have—when I can make it stick," he said dryly. "As a Med Ship man I've authority to take complete charge of any health emergency. You people have a hitherto unknown plague here. That's one emergency. The present inhabitants of the city haven't got it. That's another. So since I have authority and reason to exercise it if I can, I'm going to the city to take a little action."

"You'll be killed," said Kim.

"Possibly," admitted Calhoun. "But the number of chance happenings that could favor me is very much greater than the number of breaks that could favor the invaders. And there's the matter of colonists. Prospective colonists. You're being hunted so hard that they must be about due. They've probably

been immunized against this plague, but technically I shouldn't let them land on a plague-stricken planet."

Kim Walpole stared.

"You mean you'll try to stop them?"

"I shall try," said Calhoun, "to implement the authority vested in me by the Med Service for such cases as this. The rules about quarantine are rather strict."

"You'll be killed," said Kim, again.

Calhoun ignored the repeated prediction.

"That hunter found you," he observed, "because he knew that you'd have to drink. So he found a brook and followed it up, looking for signs of humans drinking from it. He found footprints about the spring. I found his footprints there, too. That's the trick you'll use to find other fugitives. But pass on the word not to leave tracks hereafter. For other advice, I advise you to get all the weapons you can. Modern ones, of course. You've got the blaster from the man I killed."

"I think," said Kim between his teeth, "that I'll get some more. If hunters from the city do track us to our drinking places, I'll know how to get more weapons!"

"Yes," agreed Calhoun, and added, "Murgatroyd made the antibodies that cured you. As a general rule, you can expect antibody production in your own bodies once an infection begins to be licked. In case of extreme emergency, each of you can probably supply antibodies for a fair number of other plague-victims. You might try serum from blisters you produce on your skin. Quite often antibodies turn up there. I don't guarantee it, but sometimes it works."

He paused. Kim Walpole said harshly:

"But you! Isn't there anything we can do for you?"

"I was going to ask you something," said Calhoun. He produced the telephoto films of the city as photographed from space. "There's a laboratory in the city—a biochemistry lab. Show me where to look for it."

Walpole gave explicit directions, pointing out the spot on the photo. Calhoun nodded. Then Kim said fiercely:

"But tell us something we can do! We'll be strong, presently! We'll have weapons! We'll track down-stream to where hunters leave their ground-cars and be equipped with them! We can help you!"

Calhoun nodded approvingly.

"Right. If you see the smoke of a good-sized fire in the city, and if you've got a fair number of fairly strong men with you, and if you've got ground-cars, you might investigate. But be cagey about it! Very cagey!"

"If you signal we'll come," said Kim Walpole grimly, "no matter how few we are!"

"Fine!" said Calhoun. He had no intention of calling on these weakened, starveling people for help.

He swung his depleted pack on his back again and slipped away from the glade. He made his way to the spring, which flowed clear and cool from unseen depths. He headed down the little brook which flowed away from it. Murgatroyd raced along its banks. He hated to get his paws wet. Presently, where the underbrush grew thickly close to the water's edge, Murgatroyd wailed. "*Chee! Chee!*" And Calhoun plucked him from the ground and set him on his shoulder. Murgatroyd clung blissfully there as Calhoun followed down the stream bed. He adored being carried.

Two miles down, there was another cultivated field. This one was set out to a gigantized root-crop, and Calhoun walked past shoulder-high bushes with four-inch blue-and-white flowers. He recognized the plants as of the family *solanaceae*—belladonna was still used in medicine—but he couldn't identify it until he dug up a root and found a potato. But the six-pound specimen he uncovered was still too young and green to be eaten. Murgatroyd refused to touch it.

Calhoun was ruefully considering the limitations of specialized training when he came to the end of the cultivated field. There was a highway. It was new, of course. City, fields, highways and all the appurtenances of civilized

life had been built on this planet before the arrival of the colonists who were to inhabit it. It was extraordinary to see such preparations for a population not yet on hand. But Calhoun was much more interested in the ground-car he found waiting on the highway, hard by a tiny bridge under which the brook he followed flowed.

The key he'd taken from the hunter fitted. He got in and put Murgatroyd on the seat beside him.

"These invaders, Murgatroyd," he observed, "must be in a bad way. A newly-built city which was never occupied will be like an empty house. There's no amusement or loot to be found in prowling it. They were sent to take over the planet, and they've done it. But they've nothing to do now, except hunt refugees—until their colonists arrive. I suspect they're bored. We'll try to fix that!"

He set the ground-car in motion. Toward the city.

It was a full twenty miles, but he did not encounter a single other vehicle. Presently the city lay spread out before him. He stopped and surveyed the vast pile. It was a very beautiful city. Fifty generations of architects on many worlds had played with stone and steel, groping for the perfect combination of materials with design. This city was a product of their tradition. There were towers which glittered whitely, and low buildings which seemed to nestle on the vegetation-covered ground. There were soaring bridges, and gracefully curving highways, and park areas laid out and ready. There was no monotony anywhere.

The only exception to gracefulness was the sturdy landing grid itself, half a mile high and a mile across, which was a lace-work of massive steel girders with spider-thin lines of copper woven about in the complex curves the creation of its force-field required. Inside it, Calhoun could see the ship of the invaders. It had been brought down inside the circular structure and was dwarfed by it. It gleamed there.

"And we," said Calhoun, "are going to look for a prosaic, probably messy laboratory which people who make a sport of hunting fellow-humans won't find amusing. Characters like these, Murgatroyd, aren't interested in medical science. They consider themselves conquerors. People have strange ideas!"

"*Chee!*" said Murgatroyd.

Calhoun spread out his photographs. Kim Walpole had marked where he should go and a route to it. Having been in the city while it was building, he knew even the service-lanes which, being sunken, were not a part of the city's good looks.

"But our enemies," explained Calhoun, "will not deign to use such grubby routes. They consider themselves aristocrats because they were sent as conquerors, whose job it was to clean up the dead bodies of their victims. I wonder what kind of swine are in power in the planetary government which sent them?"

He put away the photos and headed for the city again. He branched off from the rural highway where a turn-off descended into a cut. This low-level road was intended for loads of agricultural produce entering the city. It was strictly utilitarian. It ran below the surface of the park areas and entered the city without pride. It wound between rows of service-gates, behind which waste matter was some day to be assembled to be carted away for fertilizer on the fields. The city was very well designed.

Rolling along the echoing sunken road, Calhoun saw, just once, a ground-car in motion on a far-flung, cobwebby bridge between two tall towers. It was high overhead. Nobody in it would be watching grubby commerce-roads.

The whole affair was very simple indeed. Calhoun brought the car to a stop beneath the overhang of a balconied building many stories high. He got out and opened the gate. He drove the car into the cavernous, so-far-unused lower part of the building. He closed the gate behind him. He was in the center of the city, and his presence was unknown.

He climbed a new-clean flight of steps and came to the sections the public would use. There were glassy walls which changed their look as one moved between them. There were the lifts. Calhoun did not try to use them. He led Murgatroyd up the circular ramps which led upward in case of unthinkable emergency. He and Murgatroyd plodded up and up. Calhoun kept count.

On the fifth level there were signs of use, while all the others had that dusty cleanness of a structure which has been completed but not yet occupied.

"Here we are," said Calhoun cheerfully.

But he had his blaster in his hand when he opened the door of the laboratory. It was empty. He looked approvingly about as he hunted for the storeroom. It was a perfectly equipped biological laboratory, and it had been in use. Here the few doomed physicians awaiting the city's population had worked desperately against the plague. Calhoun saw the trays of cultures they'd made—dried up and dead, now. Somebody had turned over a chair. Probably when the laboratory was searched by the invaders, lest someone not of their kind remained alive in it.

He found the storeroom. Murgatroyd watched with bright eyes as he rummaged.

"Here we have the things men use to cure each other," said Calhoun oracularly. "Practically every one a poison save for its special use! Here's an assortment of spores—pathogenic organisms, Murgatroyd. One could start a plague with them. And here are drugs which are synthesized nowadays, but are descended from the compounds found on the spears of savages. Great helps in medicine. And here are the anaesthetics—poisons, too. These are what I am counting on!"

He chose, very painstakingly. Dextrethyl. Polysulfate. The one marked inflammable and dangerous. The other with the maximum permissible dose on its label, and the name of counteracting substances which would neutralize it. He burdened himself. Murgatroyd reached up a paw. Since Calhoun was carrying something, he wanted to carry something, too.

They went down the circular ramp again. Calhoun searched once more in the below-surface levels of the building. He found what he wanted—a painter's vortex gun which would throw "smoke rings" of tiny paint-droplets at a wall or object to be painted. One could vary the size of the ring at impact from a bare inch to a three-foot spread.

Calhoun cleaned the paint gun. He was meticulous about it. He filled its tank with dextrethyl brought down from the laboratory. He piled the empty containers out of sight.

"This trick," he observed, as he picked up the paint gun again, "was devised to be used on a poor devil of a lunatic who carried a bomb in his pocket for

protection against imaginary assassins. It would have devastated a quarter-mile circle, so he had to be handled gently."

He patted his pockets. He nodded.

"Now we go hunting—with an oversized atomizer loaded with dextrethyl. I've polysulfate and an injector to secure each specimen I knock over. Not too good, eh? But if I have to use a blaster I'll have failed."

He looked out a window at the sky. It was now late afternoon. He went back to the gate to the service road. He went out and piously closed it behind him. On foot, with many references to the photomaps, he began to find his way toward the landing grid. It ought to be something like the center of the invaders' location.

It was dark when he climbed other service stairs from the cellar of another building. This was the communications center of the city. It had been the key to the mopping-up process the invaders began on landing. Its call board would show which apartments had communicators in use. When such a call showed, a murder-party could be sent to take care of the caller. Even after the first night, some individual isolated folk might remain—perhaps unaware of what went on. So there would be somebody on watch, just in case a dying man called for the solace of a human voice while still he lived.

There was a man on watch. Calhoun saw a lighted room. Paint gun at the ready, he moved very silently toward it. Murgatroyd padded faithfully behind him.

Outside the door, Calhoun adjusted his curious weapon. He entered. There was a man nodding in a chair before the lifeless board. When Calhoun entered he raised his head and yawned. He turned.

Calhoun sprayed him with smoke rings—vortex rings. But the rings were spinning fissiles of vaporized dextrethyl—that anaesthetic developed from ethyl chloride some two hundred years before, and not yet bettered for its special uses. One of its properties was that the faintest whiff of its odor produced a reflex impulse to gasp. A second property was that—like the ancient ethyl chloride—it was the quickest-acting anaesthetic known.

The man by the call board saw Calhoun. His nostrils caught the odor of dextrethyl. He gasped.

He fell unconscious.

Calhoun waited patiently until the dextrethyl was out of the way. It was almost unique among vapors in that at room temperature it was lighter than air. It rose toward the ceiling. Calhoun moved forward, brought out the polysulfate injector, and bent over the unconscious man. He did not touch him otherwise.

He turned and walked out of the room with Murgatroyd piously marching behind him.

Outside, Calhoun said:

"As one medical man to another, maybe I shouldn't have done that! I doubt these invaders have a competent physician among them. But even he would be apt to think that that man had collapsed suddenly and directly into the coma of the plague. That polysulfate's an assisting anaesthetic. It's not used alone, because when you knock a man out with it he stays out for days. It's used just below the quantity that would affect a man, and then the least whiff of another anaesthetic puts him under, and he can be brought out fast and he's better off all around. But I've got this man knocked out! He'll stay unconscious for a week."

Murgatroyd piped, "*Chee!*"

"He won't die," said Calhoun grimly, "but he won't come out fighting—unless somebody wakes him earlier. And of course, he is a murderer!"

"*Chee!*" agreed Murgatroyd.

He reached up a furry paw and took hold of Calhoun's hand. They walked out into the street together.

It is notorious that the streets of a city at night are ghostly and strange. That is true of a city whose inhabitants are only asleep. There is more and worse of eeriness in a deserted city, whose inhabitants are dead. But a city which has never lived, which lies lifeless under the stars because its people never came to live in it—that has the most ghastly feel of all.

Calhoun and Murgatroyd walked hand in paw through such a place. That the invaders felt the same eeriness was presently proved. Calhoun found a place

where a light shone and voices came out into the tiny, remote night sounds of Maris III. Men were drinking in an unnecessarily small room, as if crowding together to make up for the loneliness outside. In the still night they made a pigmy tumult with their voices. They banged drunkenly on a table and on the floor.

Calhoun stood in the doorway and held the paint-gun trigger down. He traversed the room twice. Whirling rings of invisible vapor filled the place. Men gasped.

Calhoun waited a long time, because he had put a great deal of dextrethyl into a small space. But presently he went in and bent over each man in turn, while Murgatroyd watched with bright, inquisitive eyes. He arranged one figure so that it seemed to have been stricken while bending over another, fallen companion. The others he carried out, one at a time, and placed at different distances as if they had fallen while fleeing from a plague. One he carried quite a long distance, and left him with dusty knees and hands as if he had tried to crawl when strength failed him.

"They'd have been immunized at pretty well the same time, before they were shipped on this job," Calhoun told Murgatroyd. "It'll seem very plaguelike for them to fall into comas nearly together. If I found men like this, and didn't know what to do, I'd suspect that it was a delayed-action effect of some common experience—like an immunization shot. We'd better try the ship, Murgatroyd."

On the way he passed close to the control-building of the landing grid. There was a light inside it, too. There were four men on watch. Two remained inside, very, very still, when Calhoun went on. The others seemed to have fled and collapsed in the act. They breathed, to be sure. Their hearts beat solidly. But it would not be possible to rouse them to consciousness.

Calhoun didn't get into the ship, though. A chance happening intervened, which seemed an unfavorable one. Its port was locked and his cautious attempt to open it brought a challenge and a blaze of lights.



He fled for the side of the landing grid with blaster-bolts searing the ground all about him. Murgatroyd leaped and pranced with him as he ran.

VI

"The probable complete success of a human enterprise which affects non-co-operating other human beings may be said to vary inversely as the fourth power of the number of favorable happenings necessary for complete success. This formula is admittedly empirical, but its accordance with observation is remarkably close. In practice, the probability of absolute, total success in any undertaking is negligible. For this reason, mathematics and sanity alike counsel the avoidance of complex

plannings, and most especially of plans which must succeed totally to succeed at all."

Probability and Human Conduct
Fitzgerald

When morning came, Calhoun very wryly considered the situation. He couldn't know the actual state of things, to be sure. He'd been shot at. But even so—though that fact did not allow his hopes to be realized in every detail—the probability of a considerable success remained. It was not likely that the invaders would ascribe the finding of unconscious, stertorously breathing members of their number to Calhoun. Making men unconscious was not the kind of warfare a plague-refugee would use. Still more certainly, it was not what the invaders themselves would practice. To devise and spread a plague, of course, was not beyond them. That had been done. But they would not disable an enemy and leave him alive. They would murder him or nothing. So when men of their group were found in something singularly close to the terminal coma of the plague, they'd think them victims. They'd guess that their supposed immunity was only to the early symptoms, not to the final ones and death.

It should not be an encouraging opinion.

But this morning Calhoun found himself hungry. He looked remorsefully at Murgatroyd.

"I gave our rations to those refugees," he said regretfully. "I took no thought for the morrow—which has turned out to be today. I'm sorry, Murgatroyd!"

Murgatroyd said nothing.

"Maybe," suggested Calhoun, "we can find some of these invaders at a meal."

It was reckless, but recklessness was necessary in the sort of thing Calhoun had started. He and Murgatroyd ventured out into the streets. The emptiness of the city was appalling. If it had been dilapidated, if it had been partly ruined—the emptiness might have seemed somehow romantic. But every building was perfect. Each was complete but desolately unused.

Calhoun spotted a ground-car at a distance, stopped before a long, low, ground-hugging structure near the landing grid. It was perfectly suited to be

the headquarters of the strangers in the city. Calhoun considered it for a long time, peering at it from a doorway.

"We shouldn't try it," he said at last. "But we probably will. If we can make these characters so panic-stricken that they run out of the city like the earlier refugees—it would be a highly favorable happening. They might do it if their bosses were knocked out by what they thought was the plague. And besides, we should get a meal out of it. There'll be food in there."

He backtracked a long way. He darted across a road with Murgatroyd scampering beside him. He stalked the building, approaching it behind bushes, carrying the paint gun. He reached its wall. He began to crawl around the outside to reach the doorway. He heard voices as he passed the first windows.

"But I tell you we're immune!" cried a voice furiously. *"It can't be the same thing those Dettrans died from! It can't! And there was that man who ran from the ship last night—"*

Calhoun crawled on. Murgatroyd skipped. Calhoun heard an exclamation behind him. He turned his head, and Murgatroyd was fifteen feet away from the building-wall, and plainly visible to those inside. And he'd been seen fleeing with Calhoun from the ship.

Calhoun swore softly. He ran. He reached the door before which a ground-car stood. He wrenched it open and set the paint gun at work firing a steady stream of vortex rings into the interior. He drew his blaster and faced the outside world.

There was a crashing of glass. Somebody had plunged out a window. There were rushing feet inside. They'd be racing toward this doorway from within. But the hallway—anteroom—foyer—whatever was immediately inside the door would be filled with dextrethyl vapor. Men would gasp and fall.

A man did fall. Calhoun heard the crash of his body to the floor. But also a man came plunging around the building's corner, blaster out, searching for Calhoun. But he had to sight his target and then aim for it. Calhoun had only to pull trigger. He did.

Shoutings inside the building. More rushing feet. More falls. Then there was the beginning of the rasping snarl of a blaster, and then a cushioned, booming, roaring detonation which was the explosive dextrethyl vapor,

ignited by it. The blast lifted the building's roof. It shattered partitions. It blew every window out.

Calhoun sprinted for the ground-car. A blaster-bolt flashed past him. He halted and deliberately traversed the building with the trigger held down. Smoke and flame leaped up. At least one more invader crumpled. Calhoun heard a voice yelling inside somewhere.

"We're attacked! Those refugees are throwing bombs! Rally! Rally! We need help!"

It would be a broadcast call for assistance. Wherever men lolled or loafed or tried desultorily to find something to loot, they would hear it. Even the standby crew in the spaceship would hear it. Those who repaired the grid-transformers Calhoun had burned out would hear it. Men would come running. Hunters would come. Men in cars—

Calhoun snatched Murgatroyd to the seat beside him. He turned the key and the tires screamed and he shot away.

The highways were of course, superb. He raced forward, and the car's communicator began to mutter as somebody in the undamaged part of the building chattered that he'd gotten in a car and away. It described his course. It commanded that he be headed off. It hysterically demanded that he be killed, killed, killed—

Another voice took its place. This voice was curt and coldly furious. It snapped precise instructions.

Calhoun found himself on a gracefully curving, rising road. He was midway between towers when another car flashed toward him. He took his blaster in his left hand. In the split second during which the cars passed each other, he blasted it. There was a monster surge of smoke and flame as the stricken car's Duhanne cell shorted and vaporized half the metal of the car itself.



There came other voices. Somebody had sighted the explosion. The voice in the communicator roared for silence.

"You," he rasped. *"If you got him, report yourself!"*

"Chee-chee-chee!" chattered Murgatroyd excitedly.

But Calhoun did not report.

"He got one of us," raged the icy voice. *"Get ahead of him! And blast him!"*

Calhoun's car went streaking down the far side of the traffic-bridge. It rounded a curve on two wheels. It flashed between two gigantic empty buildings and came to a sideways, and plunged into that, and came again to a division and took the left-hand turn, and next time took the right. But the muttering voices continued in the communicator. A voice, by name, was ordered to the highest possible bridge from which it could watch all lower-level roadways. Others were to post themselves here, and there—and to stay

still! A group of four cars was coming out of the storage-building. Blast any single car in motion. Blast it! And report, report, report—

"I suspect," said Calhoun to the agitated Murgatroyd beside him, "that this is what is known as military tactics. If they ring us in—There aren't but so many of them, though. The trick for us is to get out of the city. We need more choices for action. So—"

The communicator panted a report of his sighting, from a cobweblike bridge at the highest point of the city. He was heading—

He changed his heading. He had so far seen but one car of his pursuers. Now he went racing along empty, curving highways, among untenanted towers and between balconied walls with blank-eyed windows gazing at him everywhere.

It was nightmarish because of the magnificence and the emptiness of the city all about him. He plunged along graceful highways, across delicately arched bridges, through crazy ramifications of its lesser traffic arteries—and he saw no motion anywhere. The wind whistled past the car windows, and the tires sang a high-pitched whine, and the sun shone down and small clouds floated tranquilly in the sky. There was no sign of life or danger anywhere on the splendid highways or in the heart-wrenchingly beautiful buildings. Only voices muttered in the communicator of the car. He'd been seen here, flashing around a steeply banked curve. He'd swerved from a waiting ambush by pure chance. He'd—

He saw green to the left. He dived down a sloping ramp toward one of the smaller park areas of the city.

And as he came from between the stone guard rails of the road, the top of the car exploded over his head. He swerved and roared into dense shrubbery, jerked Murgatroyd free despite the *tormal*'s clinging fast with all four paws and his tail, and dived into the underbrush.

He ran, swearing and plucking solidified droplets of still-hot metal from his garments and his flesh. They hurt abominably. But the man who'd fired wouldn't believe he'd missed, followed as his blasting was by the instant

wrecking of the car. The man who'd fired would report his success before he moved to view the corpse of his supposed victim. But there'd be other cars coming. At the moment it was necessary for Calhoun to get elsewhere, fast.

He heard the rushing sound of arriving cars while he panted and sweated through the foliage of the park. He reached the far side and a road, and on beyond there was a low stone wall. He knew instantly what it was. Service highways ran in cuts, now and again roofed over to hide them from sight, but now and again open to the sky for ventilation. He'd entered the city by one of them. Here was another. He swung himself over the wall and dropped. Murgatroyd recklessly and excitedly followed.

It was a long drop, and he was staggered when he landed. He heard a soft rushing noise above. A car raced past. Instants later, another.

Limping, Calhoun ran to the nearest service-gate. He entered and closed it. Scorched and aching, he climbed to the echoing upper stories of this building. Presently he looked out. His car had been wrecked in one of the smaller park areas of the city. Now there were other cars at two-hundred-yard intervals all about it. It was believed that he was in the brushwood somewhere. Besides the cars of the cordon, there were now twenty men on foot receiving orders from an authoritative figure in their midst.

They scattered. Twenty yards apart, they began to move across the park. Other men arrived and strengthened the cordon toward which he was supposed to be driven. A fly could not have escaped.

Those who marched across the park began methodically to burn it to ashes before them with their blasters.

Calhoun watched. Then he remembered something and was appalled. Among the fugitives in the glade, Kim Walpole had asked hungrily if they whose lives he had saved could not do something to help him. And he'd said that if they saw the smoke of a good-sized fire in the city they might investigate. He'd had no faintest intention of calling on them. But they might see this cloud of smoke and believed he wanted them to come and help!

"Damn!" he said wryly to Murgatroyd. "After all, there's a limit to any one series of actions with probable favorable chance consequences. I'd better start a new one. We might have whittled them down and made the

unwhittled ones run away, but I had to start using a car! And then they'd try to blame me for everything. So—we start all over with a new policy."

He explored the building quickly. He prepared his measures. He went back to the window from which he'd looked. He cracked its window.

He opened fire with his blaster. The range was long, but with the beam cut down to minimum spread he'd knocked over a satisfying number of the men below before they swarmed toward the building, sending before them a barrage of blaster-fire that shattered the windows and had the stone façade smoking furiously.

"This," said Calhoun, "is an occasion where we have to change their advantage in numbers and weapons into an unfavorable circumstance for them. They'll be brave because they're many. Let's go!"

He met the two ground-car loads of refugees with his arms in the air. He did not want to be shot down by mistake. He said hurriedly, when Kim and the other lean survivors gathered about him:

"Everything's all right. We've a pack of prisoners but we won't bother to feed them intravenously for the moment. How'd you get the ground-cars?"

"Hunters," said Kim savagely. "We found them and killed them and took their cars. We found some other refugees, too, and I cured them—at least they will be cured. When we saw the smoke we started for the city. Some of us still have the plague, but we've all had our serum shots." His face worked. "When we started for the city, another car overtook us. Naturally he wasn't suspicious of a car! We blasted him. Half of us have arms, now."

"I don't think we need them," said Calhoun. "Our prisoners are quite peacefully sleeping. They stormed a building where I'd fired on them, and I'd dumped some dextrethyl in the air-conditioning system. They keeled over. Later, Murgatroyd and I went in and made their slumber more ... ah ... lasting with polysulfate. The few who weren't caught were ... ah ... demoralized. I think the city's clean, now. But we've got to get to the landing-grid control room. There are some calls coming in from space. I

think the first shipload of colonists is arriving. I didn't answer, so they went in orbit around the planet. I want you people to talk to them."

"We'll bring their ship down," said the bearded man hungrily, "and blast them as they come out the exit port!"

Calhoun shook his head.

"To the contrary," he said regretfully. "You'll put on the clothes of some of our prisoners. You'll tell the arriving colonists that the plague hit you, too. You'll pretend to be one of the characters we really have safely sleeping, and you'll say all the rest have been bowled over by the plague that was sowed here to win the planet for the characters you're talking to. If they land, they'll die—or so you'll tell them. And so they will all go home, very unhappy, and they'll tell the public about it. And there will be no more shiploads of colonists arriving. We don't want them. If we persuade them to go home and not come back, there are fewer chances of unfavorable consequences to us."

The bearded man growled. But later he was one of the most convincing of the scarecrow figures whose images appeared in the vision-plates of the ship overhead. He was especially pathetic and alarming. When he'd finished, there'd have been a mass mutiny of the passengers had the spaceship skipper tried to land them.

Later, all the fugitives were very conscientious about bringing the captive invaders out of the lethargy that had been begun by dextrethyl and reinforced by polysulfate. They enjoyed their labor, after Calhoun explained.

"They came in their own ship," he said mildly, "and it's still in the landing grid—which they repaired for us, by the way. And I've been aboard the ship with Kim, here, and we've smashed their drive and communicators, and wrecked their Duhanne circuits. We took out the breech-plugs of their rockets and dumped their rocket fuel. Of course we removed their landing boats. So we're going to put them in their ship and hoist them up to space with the landing grid, and we're going to set them in a lovely orbit, to wait until we've time to spare for them. Up there they can't run or land or even signal if another shipload of colonists turns up. They'll feed themselves and they won't need guarding, and they'll be quite safe until we get help from Dettra. And that will come as soon as the Med Service has told Dettra that it wasn't a plague but an invasion that seemed to take their colony away from them."

"But—" That was Kim Walpole, frowning.

"I'm bringing my ship to the grid," said Calhoun, "and we'll recharge my Duhanne cells and replace my vision screens. I can make it here on rocket power, but it's a long way back to Headquarters. So I'll report, and a field team will come here from Med Service to get the exact data on the plague, and just how the synergy factor worked, and to make everything safe for the people the city was built for. Incidentally, I've a tiny blood-sample from Helen that they can get to work on for the bacteriology."

Kim said, frowning:

"I wish we could do something for you!"

"Put up a statue," said Calhoun dryly, "and in twenty years nobody will know what it was for. You and Helen are going to be married, aren't you?" When Kim nodded, Calhoun said, "In course of time, if you remember and think it worth while, you may inflict a child with my name. That child will wonder why, and ask, and so my memory will be kept green for a full generation!"

"Longer than that!" insisted Kim. "You'll never be forgotten here!"

Calhoun grinned at him.

Three days later, which was six days longer than he'd expected to be aground on Maris III, the landing grid heaved the little Med Ship out to space. The beautiful, nearly-empty city dwindled as the grid-field took the tiny spacecraft out to five planetary diameters and there released it. And Calhoun spun the Med Ship about and oriented it carefully for that place in the Cephis cluster where Med Service Headquarters was, and threw the overdrive switch.

The universe reeled. Calhoun's stomach seemed to turn over twice, and he had a sickish feeling of spiraling dizzily in what was somehow a cone. He swallowed. Murgatroyd made gulping noises. There was no longer a universe preceptible about the ship. There was dead silence. Then those small random noises began which have to be provided if a man is not to

crack up in the dead stillness of a ship traveling at thirty times the speed of light.

Then there was nothing more to do. In overdrive travel there is never anything to do but pass the time away.

Murgatroyd took his right-hand whiskers in his right paw and licked them elaborately. He did the same to his left-hand whiskers. He contemplated the cabin, deciding upon a soft place in which to go to sleep.

"Murgatroyd," said Calhoun severely, "I have to have an argument with you! You imitate us humans too much! Kim Walpole caught you prowling around with an injector, starting to give our prisoners another shot of polysulfate! It might have killed them! Personally, I think it would have been a good idea, but in a medical man it would have been most unethical. We professional men have to curb our impulses! Understand?"

"*Chee!*" said Murgatroyd. He curled up and wrapped his tail meticulously about his nose, preparing to doze.

Calhoun settled himself comfortably in his bunk. He picked up a book. It was Fitzgerald on "Probability and Human Conduct."

He began to read as the ship went on through emptiness.

THE END

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